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Falcon 9 CRS-10 Vehicle Simulator and Navigation

Topics in Aerospace Engineering - Team 5

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Contents

1 Introduction and Project Scope 2

2 Simulation Architecture and Shared Conventions 3

3 Vehicle Configuration, Dynamics, and Mission Definition 5

4 Environment Models 10

5 Aerodynamics 12

6 Sensor Models and Measurement Generation 14

7 Navigation Architecture and State Estimation 16

8 Mission Execution and Numerical Implementation 18

9 Verification, Validation, and Simulation Results 20

10 Engineering Assessment and Implications for Future GNC 26

11 Conclusions 27

References 28

A Reproducibility and Supporting Data 29

1 Introduction and Project Scope

1.1 Project 2 objectives

Project 2 develops a physics-based Falcon 9 simulator and navigation driven by simulated sensor measurements. Forces, moments, changing mass properties and numerical integration generate motion; imperfect observations of that motion drive state estimation. The course defines this environment as the computational foundation for guidance in Project 3 and integrated control in Project 4 [11, slides 2–4].

The required models cover equations of motion, coordinate systems, gravity, atmosphere, propulsion, aerodynamics, mass/inertia variation and configuration changes. Navigation addresses position, velocity, attitude and angular velocity. Evaluation therefore separates model verification, comparison with public flight evidence, and estimation error relative to simulation truth [11, slides 4–6].

1.2 Why CRS-10?

CRS-10 is SpaceX's tenth contracted Dragon cargo-resupply mission to the ISS. Official material identifies LC-39A launch, first-stage recovery at LZ-1, and approximate cutoff, separation and restart times [1]. Its webcast and published ascent digitization provide a quantitative reference [13, 14]. Common-stack ascent, booster return and upper-stage continuation exercise both continuous propagation and configuration changes.

The modeled payload is one loaded Dragon/trunk assembly, not a satellite and separate fairing. NASA's 2490 kg cargo inventory is included within that assembly, not its total mass [2]. Figure 1 identifies the hardware and official mission document; Section 3 qualifies the mixed-source internal properties.



Figure 1 CRS-10 reference configuration. Left: Falcon 9 and Dragon at LC-39A before launch, NASA/Glenn Benson [12]. Right: header and mission-patch excerpts from the official SpaceX press kit [1]. The images identify the hardware and source; they do not determine the internal mass distribution.

1.3 Demonstration and validation scope

Both children inherit the propagated separation state. The booster returns to geometric contact, while the upper-stage/Dragon child continues under assumed commands. First-stage navigation uses gyro-propagated attitude and GNSS/LiDAR updates with a gated star-tracker interface. Truth supplies virtual observations, assumed initial alignment and evaluation, rather than continuous estimator attitude input.

External comparison covers approximately 0–140 s of E-class digitized ascent readouts. The ascent operating point and pitch timescale were tuned against this same record; their agreement is calibration evidence, not independent validation. The later chapters retain that distinction when reporting flight, contact and navigation results. Return robustness, reconstructed SpaceX guidance and validated orbital insertion are outside the demonstrated scope.

2 Simulation Architecture and Shared Conventions

2.1 Physical truth, measurements, and estimates

The physical branch computes loads and mass properties and integrates vehicle motion. The measurement branch samples that truth through sensor error and availability models. Navigation propagates its own attitude and estimates the IMU-station motion. Comparison uses the same station and time. Figure 2 separates these interfaces.

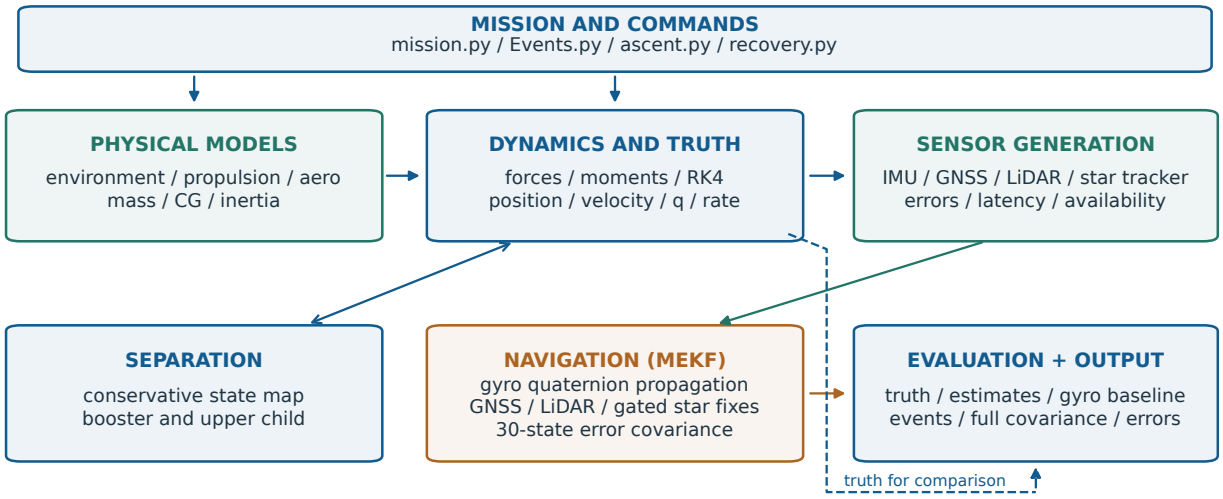


Figure 2 Implemented data flow. The MEKF uses simulated sensor measurements and its own propagated attitude. Truth supplies sensor generation and evaluation; initial alignment is assumed. Return commands use physical-state feedback, while navigation is evaluated separately.

The mission layer selects the initial configuration, commands, and accepted phase. A throttle command becomes thrust and flow through propulsion; an aerodynamic configuration and fin command become force and moment through aerodynamics. The dynamics model then calculates the resulting motion. Neither the force provider nor the mass model prescribes the reference altitude/speed history. Table 1 assigns the corresponding implementation responsibilities.

Function	Main modules	Interface

Vehicle properties	`VehicleData.py`
	`vehicle_mass_model.py`
	`mass_properties.py`
		Trial inventory to mass, CG, inertia, and derivatives.

Environment and loads	`Environement.py`
	`propulsion.py`
	`Aerodynamics.py`
	`load_interface.py`
		Trial state/commands to loads and exhaust data.

Motion and configuration	`dynamics.py`
	`frames.py`
	`staging.py`
		Derivative, integrated state, and inherited child states.

Measurement and estimation	`sensors.py` `kalman.py`	Sensor-station truth to observations and separate estimates.

Mission execution	`main.py`
	`mission.py`
	`Events.py`
		Commands, accepted clocks/events, state synchronization, and saved output.

Table 1 Responsibilities of the shared implementation. Configuration and scenario decisions are kept separate from reusable component arithmetic and equations of motion.

2.2 Frames, origins, and attitude convention

Frame B is right-handed and fixed to the vehicle, with $+x_B$ toward the nose. Frame E is Earth-centred, launch-aligned, and rotating with Earth. Code identifiers call it ECEF, but its axes are not conventional Greenwich-axis

ECEF. At the launch site, $+x_E$ is north, $-y_E$ east, and $+z_E$ up; the initial structural base is $[0, 0, R_E]^\mathsf{T}$ with $R_E = 6371000\,\mathrm{m}$.

The scalar-first Hamilton quaternion $\boldsymbol{q}_{BE} = [q_w, q_x, q_y, q_z]^\mathsf{T}$ maps Body components into E using $\mathbf{C}_{BE}$. Equation (1) defines the vector transformation and launch orientation:

$$\boldsymbol{u}^E = \mathbf{C}_{BE}\boldsymbol{u}^B, \qquad \boldsymbol{q}_{BE}(0) = \begin{bmatrix} \sqrt{1/2} & 0 & -\sqrt{1/2} & 0 \end{bmatrix}^\mathsf{T}, \qquad \mathbf{C}_{BE}(0) = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}. \tag{1}$$

Thus Body $+x$ is initially upward, Body $+y$ maps to $+y_E$, and Body $+z$ maps to $-x_E$. Quaternions are normalized, and $\boldsymbol{q}$ and $-\boldsymbol{q}$ denote the same orientation. The adopted latitude $\lambda = 28.6°$ rotates the Earth-rate vector into the launch-aligned frame as $\boldsymbol{\Omega}^E = \Omega_E[\cos\lambda, 0, \sin\lambda]^\mathsf{T}$, with $\Omega_E = 7.292115 \times 10^{-5}\,\mathrm{rad\,s^{-1}}$. It does not relocate the pad into Greenwich-axis coordinates.

Reference point and component basis are separate choices. Table 2 distinguishes them. A structural station $\boldsymbol{s}^B$ is measured from the stage base, whereas the corresponding CG force arm is $\boldsymbol{d}^B = \boldsymbol{s}^B - \boldsymbol{c}^B$, where $\boldsymbol{c}^B$ is the current CG station.

Quantity	Meaning
Dynamic position $\boldsymbol{r}_C^B$	Earth centre to instantaneous CG, expressed in current Body axes.
Dynamic velocity $\boldsymbol{v}_C^B$	CG velocity relative to rotating Earth, in current Body components.
Vehicle-object position/velocity	Structural-base state, with the same external origin and Body component basis.
Component/sensor station $\boldsymbol{s}^B$	Fixed structural location measured from the relevant stage base.
Moment and inertia	About the instantaneous vehicle CG, in Body axes.
Filter position/velocity	At the fixed IMU station, expressed in frame E.

Table 2 Shared origins and representations. Storing an Earth-centred vector in Body components does not make its origin the CG.

2.3 State and load interfaces

The physical state stores 15 numbers: CG position and velocity in Body components, a four-component quaternion, Body angular rate, and two stage-indexed propellant inventories. Each child retains this layout after separation, with the detached inventory set to zero. The estimator instead tracks the fixed IMU station in E. Its nominal/error-state definitions belong to Section 7; sensor stations and rate conventions belong to Section 6. Comparisons always use matching points, frames and epochs.

The load record separates nongravitational force, CG moment, central gravity, rotating-frame acceleration, stage consumption and nozzle-exit flow data. A station force contributes $(\boldsymbol{s} - \boldsymbol{c}) \times \boldsymbol{F}$ once; free moments are supplied separately. Gravity and frame acceleration enter the dynamics once as accelerations. This contract lets environment, mass, dynamics and navigation models evolve without redefining shared variables.

Trial evaluations recompute loads and properties without advancing accepted inventories, event clocks or sensor random generators. Only accepted states produce measurements and output. Section 8 describes the execution order, step boundaries and child propagation; Section 9 evaluates the resulting flight and navigation histories. Return commands currently use physical-state feedback, so landing performance and navigation accuracy are separate experiments.

3 Vehicle Configuration, Dynamics, and Mission Definition

3.1 Configuration and evidence-qualified parameters

The attached vehicle comprises stage 1, stage 2, and a loaded Dragon/trunk assembly. After separation, the lower vehicle contains stage 1 and the upper vehicle contains stage 2 and Dragon. The parameter register distinguishes K (authoritative published information), E (literature/data-derived estimate), and A (explicit modeling assumption). E/A denotes a sourced estimate whose use also requires an assumption. Source authority and applicability to the selected 2017 flight are separate considerations. Appendix A.1 consolidates the parameter/source register; this chapter develops the governing model.

Table 3 consolidates the configuration used by `vehicle_parameters.py` and `VehicleData.py`. The 62.4 m equivalent height is the sum of adopted stage/Dragon lengths; it is not presented as a measured CRS-10 overall dimension. Stage boundaries and the contiguous attachment datums remain conditional. The 8430 kg loaded Dragon mass is retained as a provisional external estimate; the NASA cargo inventory is included within it. No separate satellite fairing is added.

Parameter	Adopted value	Class	Basis
Outer diameter	3.66 m	K	Later official external geometry [5].
Stage-1 / stage-2 length	42.6/12.6 m	E/A	Legacy FT geometry and assumed station allocation [17].
Stage-1 dry / propellant mass	22200/410900 kg	E	Adopted research mass budget [7].
Stage-2 dry / propellant mass	4000/107500 kg	E	Same literature set [7].
Loaded Dragon assembly	8430 kg	E	Provisional external estimate [3].
Dragon equivalent length / diameter	7.2/3.66 m	E/A	Earlier Dragon geometry transferred to a uniform assembly [4].
Dragon / stage-2 CG station	58.8/48.9 m	A	Midpoints above assumed attached-stage datums.
Stage-1 nominal dry CG	17.521 m	E/A	Component budget and residual allocation.
Initial attached mass	553030 kg	E/A	Sum of the adopted inventories and structures.
Equivalent model height	62.4 m	E/A	Sum of the adopted axial envelopes.

Table 3 Consolidated vehicle inputs. Stations are measured from the attached stack base along Body $+x$. A known external diameter does not establish a known internal tank or mass distribution.

3.2 Propulsion and propellant depletion

Stage 1 has one central engine and eight surrounding engines on an assumed 1.3 m radius ring. Stage 2 has one MVac engine. The force-application pivots are 1.0 m and 0.5 m above the respective bases (A); pivot stations, engine mass-centre stations, and nozzle-exit stations are distinct. Engine dry masses are already included in the stage dry budgets.

For a running steady engine with throttle u, the implementation evaluates Equation (2); an off engine has zero thrust and flow:

$$T = uT_{\mathrm{vac}} - A_e p_{\mathrm{amb}}, \qquad Q = \frac{uT_{\mathrm{vac}}}{I_{\mathrm{sp,vac}} g_0}, \qquad g_0 = 9.80665\,\mathrm{m\,s^{-2}}. \tag{2}$$

Here T is thrust, Q positive propellant outflow, A_e nozzle exit area, p_{amb} ambient pressure, and $I_{\mathrm{sp,vac}}$ the adopted vacuum specific impulse. The code retains a fixed-area ambient-pressure correction; the steady throttle does not scale that pressure term. Smooth ignition/shutdown envelopes subsequently scale the evaluated thrust and flow together. The effective stage-1 vacuum point is $756000 + 0.68(101325) = 824901$ N at full throttle, using the historical sea-level reference and a literature exit area. This mixed-source operating set is E/A, not a certified CRS-10 engine rating [6, 7]. Table 4 lists the adopted engine inputs.

Parameter	Stage-1 engine	Stage-2 engine
Engine count	9	1
Adopted vacuum thrust	824.901 kN (E/A)	935 kN (E)
Vacuum specific impulse	311 s (E)	348 s (E)
Exit area	0.68 m^2 (E)	4.90 m^2 (E)
Steady minimum throttle	0.60 (E)	0.40 (E)

Table 4 Engine operating inputs [7, 6]. The numerical scenario chooses engine count and throttle separately from these engine relations. A stored 5° gimbal parameter (A) does not establish implemented physical gimbal dynamics.

Stage-1 loading is $m_{f,0} = 123500\,\text{kg}$ of RP-1 and $m_{o,0} = 287400\,\text{kg}$ of LOX [9]. Their loaded ratio is 2.32713, while consumption uses $\eta = Q_o/Q_f = 2.35$ [8]. Equation (3) reconstructs both tanks from the stage's total consumed mass Δm:

$$Q_f = \frac{Q}{1+\eta}, \quad Q_o = \frac{\eta Q}{1+\eta}, \qquad m_f = m_{f,0} - \frac{\Delta m}{1+\eta}, \quad m_o = m_{o,0} - \frac{\eta \Delta m}{1+\eta}. \tag{3}$$

The adopted combination depletes LOX first and leaves 1202.128 kg of RP-1 at its mathematical limit. That residual is unusable under the assumed fixed ratio and is separate from a mission reserve. Reconstruction assumes fixed initial loading and no venting, refilling, or variable mixture schedule. Stage 2 retains an equivalent uniformly distributed propellant cylinder rather than the first-stage dual-column model.

3.3 Changing mass, CG, and inertia

3.3.1 Liquid and dry-component models

Each first-stage liquid is a settled, flat-ended cylindrical column at constant density. Table 5 lists its geometry and reference conditions. Structural section lengths are transferred to equivalent tank envelopes; domes, internal hardware displacement, and datum alignment are assumptions.

Quantity	RP-1	LOX
Inner radius [m], E/A	1.8269	1.8271
Bottom station [m], E/A	4.0	18.7
Equivalent section length [m], E/A	14.7	22.1
Density [kg m^{-3}], E	820	1250
Reference temperature [K], E	266	66

Table 5 Equivalent tank inputs. Radii subtract the modeled skin thickness from 1.83 m; section lengths/stations use Ecker's structural model [15, Table 1]. Density/temperature pairs use Miquel's Table 5, whose reference pressure is unspecified [16].

For liquid mass m_i, density ρ_i, radius r_i, bottom station b_i, and outflow Q_i, Equation (4) defines height, centroid, and centroid rate:

$$h_i = \frac{m_i}{\rho_i \pi r_i^2}, \qquad c_i = b_i + \frac{h_i}{2}, \qquad \dot{c}_i = -\frac{Q_i}{2\rho_i \pi r_i^2}. \tag{4}$$

The height must remain inside the adopted envelope. Slosh, ullage motion, reverse-acceleration settling, and feed transients are not resolved. The settled-column approximation is retained through coast and return; restart settling is a separate scenario assumption.

The dry allocation is shell 8000 kg at 21.0 m, fins 1520 kg at 40.3 m, engines 4230 kg at 0.452 m, and folded legs 2000 kg at 6.66 m, using published group estimates with equivalent shapes [8, Table 2]. The remaining 6450 kg closes the 22200 kg budget and is assumed uniformly distributed over 4.0–40.8 m. This yields nominal dry CG 17.521 m. Moving the residual group's centroid anywhere within that interval gives conditional dry-CG bounds 12.18–22.87 m; they are allocation bounds, not statistical confidence limits.

The shell/residual groups use thin-shell equivalents and localized groups use equivalent disks. The liquid columns and loaded Dragon use uniform solid cylinders. Dry-group placements stay fixed; deployed aerodynamic/contact geometry does not separately relocate the dry leg mass. The initial combined vehicle has CG

station 29.3704 m and inertia diagonal approximately $[9.47993 \times 10^5, 1.13857 \times 10^8, 1.13857 \times 10^8]\,\mathrm{kg\,m^2}$. These are model outputs (E/A), not measured CRS-10 mass properties.

3.3.2 Component aggregation and analytical derivatives

A `MassComponent` describes one part or liquid about its own centre. `MassProperties` describes the attached vehicle about its current CG. With component mass m_i, axial centroid c_i, local inertia $\mathbf{J}_{i,c}$, and offset $\boldsymbol{d}_i = (c_i - c)\boldsymbol{e}_x$, Equation (5) gives the combined properties:

$$\begin{aligned} m &= \sum_i m_i, \qquad c = \frac{\sum_i m_i c_i}{m}, \\ \mathbf{J}_C &= \sum_i \left[\mathbf{J}_{i,c} + m_i \mathbf{S}(\boldsymbol{d}_i)\right], \qquad \mathbf{S}(\boldsymbol{d}) = (\boldsymbol{d}^\top \boldsymbol{d})\mathbf{I} - \boldsymbol{d}\boldsymbol{d}^\top. \end{aligned} \tag{5}$$

Here $\mathbf{I}$ is the identity matrix. The implementation differentiates both component properties and parallel-axis offsets. Equation (6) states the first derivatives; second CG/mass derivatives are also evaluated analytically for the motion and sensor interfaces:

$$\begin{aligned} \dot{c} &= \frac{\sum_i (\dot{m}_i c_i + m_i \dot{c}_i) - \dot{m} c}{m}, \\ \dot{\mathbf{J}}_C &= \sum_i \left[\dot{\mathbf{J}}_{i,c} + \dot{m}_i \mathbf{S}(\boldsymbol{d}_i) + m_i \dot{\mathbf{S}}(\boldsymbol{d}_i)\right]. \end{aligned} \tag{6}$$

Axial placement makes the parallel-axis addition $\mathrm{diag}(0, d_i^2, d_i^2)$. Thus Body I_{xx} is axial and I_{yy}, I_{zz} are transverse. Figure 3 shows this same operation in code, and Figure 4 shows its first-stage-only inventory dependence. The late change in CG trend follows the changing weights of the two liquid centroids and the fixed dry allocation.

mass_properties.py | lines 157-169

```
157     inertia = np.zeros((3, 3))
158     inertia_rate = np.zeros((3, 3))
159     for c in components:
160         distance = c.center_axial_m - center
161         distance_rate = c.center_rate_mps - center_rate
162         parallel = np.diag([0.0, distance**2, distance**2])
163         parallel_rate = np.diag([0.0] + [2 * distance * distance_rate] * 2)
164         inertia += c.inertia_at_center_kgm2 + c.mass_kg * parallel
165         inertia_rate += (
166             c.inertia_rate_at_center_kgm2ps
167             + c.mass_rate_kgps * parallel
168             + c.mass_kg * parallel_rate
169         )
```

Figure 3 Exact parallel-axis and derivative accumulation in `mass_properties.py`. Each local inertia is shifted to the combined CG once. Source line numbers and file hash are retained with the figure provenance.

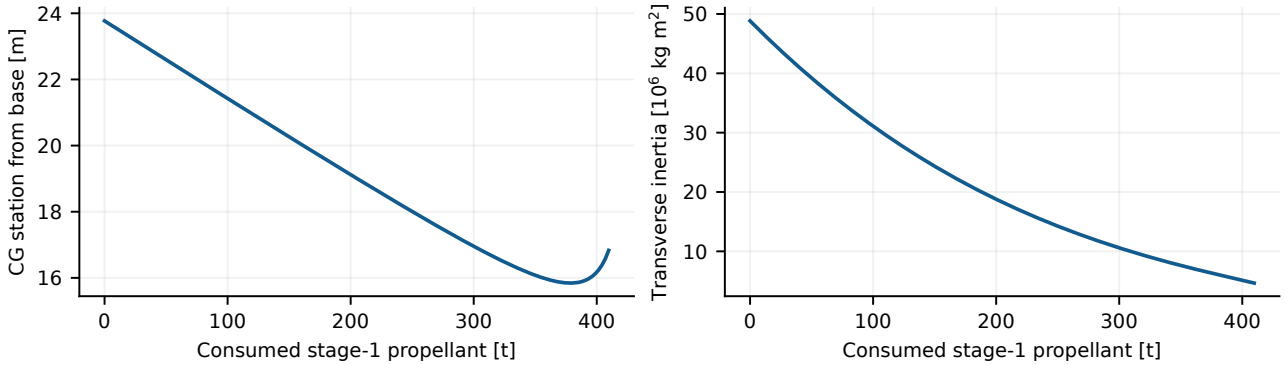


Figure 4 Stage-1-only mass-property sweep from full loading to the fixed-ratio limiting inventory. The curves evaluate the adopted component model against consumed propellant; they are neither measured flight properties nor a separate trajectory simulation.

3.4 Equations of motion and load closure

The changing-mass model uses fixed nozzle-exit stations and balanced transverse outflow. For outlet j, $Q_j \geq 0$ is outward flow, $\mu_j = -Q_j$ its signed mass-removal rate, $\dot{\mu}_j = -\dot{Q}_j$, and $\boldsymbol{d}_j = \boldsymbol{s}_j - \boldsymbol{c}$ the exit arm. $\mathbf{G}_j$ is the outlet's second-moment-per-unit-mass tensor in m^2. The open-system balance retains explicit flow terms, consistent with control-volume rocket dynamics [18].

Equations (7) and (8) are the implemented CG balances. $\boldsymbol{a}_C^B$ is the external-frame coordinate acceleration expressed in Body components; $\boldsymbol{g}^B$ and $\boldsymbol{a}_{\mathrm{rot}}^B$ are supplied separately by the environment:

$$m\boldsymbol{a}_C^B = \boldsymbol{F}^B + m\boldsymbol{g}^B + m\boldsymbol{a}_{\mathrm{rot}}^B + \sum_j \left[2\mu_j(\boldsymbol{\omega}^B \times \boldsymbol{d}_j^B - \dot{\boldsymbol{c}}^B) + \dot{\mu}_j \boldsymbol{d}_j^B\right], \tag{7}$$

$$\mathbf{J}_C\dot{\boldsymbol{\omega}}^B = \boldsymbol{M}_C^B - \boldsymbol{\omega}^B \times (\mathbf{J}_C\boldsymbol{\omega}^B) - \dot{\mathbf{J}}_C\boldsymbol{\omega}^B + \sum_j \mu_j[\mathbf{S}(\boldsymbol{d}_j^B) + \mathbf{G}_j]\boldsymbol{\omega}^B. \tag{8}$$

Pressure-corrected thrust already contains the exhaust reaction; another Qv_e force is not added. The remaining flow terms describe reference migration and angular-momentum outflow. The current closure requires axial CG migration, fixed exits, and balanced transverse exhaust centroids; internal swirl and moving nozzle boundaries are omitted. The code verifies consistency between consumption and mass derivatives before evaluating these balances.

Because the state is stored in current Body axes, Equation (9) includes basis-transport terms. The quaternion uses the same approximate relative-rate convention as Section 2:

$$\begin{aligned} \dot{\boldsymbol{r}}_C^B &= \boldsymbol{v}_C^B - \boldsymbol{\omega}^B \times \boldsymbol{r}_C^B, & \dot{\boldsymbol{v}}_C^B &= \boldsymbol{a}_C^B - \boldsymbol{\omega}^B \times \boldsymbol{v}_C^B, \\ \dot{\boldsymbol{q}}_{BE} &= \tfrac{1}{2}\boldsymbol{q}_{BE} \otimes [0, (\boldsymbol{\omega}^B)^{\mathsf{T}}]^{\mathsf{T}}, & \dot{m}_{p,k} &= -Q_k. \end{aligned} \tag{9}$$

The integrator temporarily uses external translational components within RK4, then returns the accepted state in Body components. This avoids artificial radial interpolation error from rotating an Earth-scale stored vector.

Structural-base and CG states are related by Equation (10), with base O and current CG station $\boldsymbol{c}^B$:

$$\boldsymbol{r}_C^B = \boldsymbol{r}_O^B + \boldsymbol{c}^B, \qquad \boldsymbol{v}_C^B = \boldsymbol{v}_O^B + \boldsymbol{\omega}^B \times \boldsymbol{c}^B + \dot{\boldsymbol{c}}^B. \tag{10}$$

The same transformation applies to a fixed sensor station after replacing the base-to-CG displacement appropriately. `to_state` and `apply_state` use matching flow/rate inputs. Their initial stationary-base condition can therefore have a nonzero geometric-CG velocity from depletion, without implying motion of the sensor mount.

Control moments are supplied inputs to these equations. Their idealized authority and physical actuator limitations are defined in Section 8.

3.5 Initial state and flight-phase definition

The structural base starts at the launch location of Section 2, with zero velocity relative to rotating Earth and zero Body-relative angular rate. The launch quaternion sets Body $+x$ upward; inventories are the full values in Table 3. Commands, configuration transitions, and continuous integration then generate all subsequent states.

Table 6 defines the mission sequence without substituting simulated event times for input conditions. The official schedule labels its entries approximate. Adopting a published time as a simulation command is an A-class choice even when the published schedule itself is authoritative [1].

First contact uses four assumed leg tips 1.0 m below the base at 3.5 m radius. A successful contact additionally requires deployed legs, vertical speed at most $2\,\mathrm{m\,s^{-1}}$, horizontal speed at most $1\,\mathrm{m\,s^{-1}}$, tilt at most 5°, angular-rate magnitude at most $1\,\mathrm{deg\,s^{-1}}$, and target error at most 10 m. These are A-class project limits, not certified Falcon 9 requirements. The target is an assumed point 14 km south and 6 km east of launch on the ground sphere, rather than surveyed LZ-1 coordinates.

For normalized ramp time $\xi = (t - t_s)/\Delta t$, Equation (11) supplies a smooth shutdown amplitude and its time derivative on $0 \leq \xi \leq 1$:

$$\gamma = 1 - 10\xi^3 + 15\xi^4 - 6\xi^5, \qquad \dot{\gamma} = -\frac{30\xi^2(1-\xi)^2}{\Delta t}, \qquad \Delta t = 1\,\mathrm{s}. \tag{11}$$

The envelope is constant outside the ramp, with zero endpoint slope and curvature. Scaling thrust and flow together while supplying the analytical flow derivative avoids an unmodeled discontinuous-flow impulse. The duration is a numerical assumption, not a measured engine transient. Return planning budgets, steering laws, and actuator allocation are kept in mission execution rather than embedded in the vehicle equations.

Transition	Definition in the selected scenario
Liftoff / ascent	Initial attached vehicle; prescribed ascent commands.
Shutdown begins	140 s; project one-second smooth ramp.
MECO / stack coast	141 s; thrust and flow reach zero.
Stage separation	144 s; attached parent, engines off, zero flow and flow derivative.
Upper-stage ignition	152 s; independent upper-child commands.
Boostback ignition	161 s; detached booster and fuel/attitude/readiness guards.
Boostback end / return coast	Miss, fuel, or duration condition followed by smooth shutdown.
Entry ignition	Descending through 45 km with restart guards.
Entry end / aerodynamic descent	$500\,\mathrm{m\,s^{-1}}$ speed target or planning cap, then shutdown.
Landing ignition	Project stopping-clearance prediction with restart guards.
First contact	Lowest assumed deployed leg tip reaches the ground sphere; evaluate landing criteria.
Upper-stage cutoff	545 s; prescribed upper-child shutdown.

Table 6 Physical phase definitions and command references. Entry/landing conditions and return planning criteria are project assumptions. Their actual execution times and numerical results belong to the simulation-results analysis.

3.6 Separation and inherited child states

The separation map assumes zero impulse and no discarded mass. Both children inherit the parent's attitude and angular rate. For a cut offset $\boldsymbol{\ell}^B$ from the parent structural base, the lower base retains the parent's base state and the upper base follows Equation (12):

$$\boldsymbol{r}^B_{O,u} = \boldsymbol{r}^B_{O,p} + \boldsymbol{\ell}^B, \qquad \boldsymbol{v}^B_{O,u} = \boldsymbol{v}^B_{O,p} + \boldsymbol{\omega}^B \times \boldsymbol{\ell}^B. \tag{12}$$

Upper-child component stations are rebased consistently, while Dragon stays on that child. Different child CG coordinates are expected because the tracked points have changed. Position/velocity continuity is checked at fixed structural points and at the recombined mass centre.

The conservation audit uses angular momentum about the parent CG. With child CG offset $\boldsymbol{\rho}_k$ and velocity difference $\Delta\boldsymbol{v}_k$ from the parent CG, Equation (13) includes spin and displaced-CG motion:

$$\boldsymbol{H}_{\text{after}} = \sum_k \left[\mathbf{J}_k\boldsymbol{\omega}_k + \boldsymbol{\rho}_k \times (m_k\Delta\boldsymbol{v}_k)\right]. \tag{13}$$

The map compares this value with the parent spin angular momentum, and also checks mass, linear momentum, kinetic energy, recombined CG, and structural-point continuity. Relative conservation tolerance is 10^{-10}; position and velocity continuity tolerances are $10^{-8}\,\mathrm{m}$ and $10^{-8}\,\mathrm{m\,s^{-1}}$; attitude/rate checks use 10^{-12} in their respective norms. These are numerical consistency criteria for the assumed map, not physical separation accuracy.

Afterward, stage 1 excludes stage 2 and Dragon from its mass properties, and the upper child excludes stage 1. Both use inherited states and separate subsequent commands on the same clock. The upper stage retains the uniform propellant/Dragon approximations and its separate drag provider, rather than automatically selecting the declared `second_stack` aerodynamic configuration. Dragon remains attached over the 600 s demonstration interval. The sequence tests continuous configuration/state handling; public validation of the complete return and upper-stage command laws is outside the available ascent reference.

4 Environment Models

The environment module (`Environement.py`) is a pure service layer: it supplies the world the vehicle flies through and never calls back into vehicle, dynamics or control code. Its services are the standard atmosphere, point-mass gravity with its gradient, the acceleration of the rotating Earth-fixed frame, the wind, and the ground reference; all evaluations are stateless functions of the current state. Following the project convention, any state outside a modeled domain is rejected with an error, never extrapolated.

4.1 Atmosphere

The atmosphere follows the US Standard Atmosphere 1976 [10]: seven gradient layers between sea level and the mesopause at 84.852 km, reproduced layer by layer. Within a layer with base temperature T_b, base pressure p_b and lapse rate L, and in the isothermal case, the pressure is given by Equation (14):

$$p(h) = p_b \left(\frac{T_b + L\,(h - h_b)}{T_b} \right)^{-\frac{g_0}{R\,L}} \quad (L \neq 0), \qquad p(h) = p_b \, \exp\left(-\frac{g_0\,(h - h_b)}{R\,T_b} \right) \quad (L = 0), \tag{14}$$

with $g_0 = 9.80665\,\mathrm{m/s^2}$ and the gas constant of dry air $R = 287.05287\,\mathrm{J/(kg\,K)}$. Density and speed of sound follow the ideal-gas and isentropic relations, $\rho = p/(R\,T)$ and $a = \sqrt{\gamma R T}$ with $\gamma = 1.4$. Sea-level values are $T = 288.15\,\mathrm{K}$, $p = 101\,325\,\mathrm{Pa}$, $\rho = 1.225\,\mathrm{kg/m^3}$.

Above the mesopause the model deliberately stops claiming physics: an isothermal tail at $T_m = 186.946\,\mathrm{K}$ decays exponentially with a scale height of $R\,T_m/g_0 \approx 5.47\,\mathrm{km}$ toward numerical vacuum (density below $10^{-150}\,\mathrm{kg/m^3}$ at the 2 000 km envelope ceiling; $p(200\,\mathrm{km}) \approx 2.7 \times 10^{-10}\,\mathrm{Pa}$). The envelope 0–2 000 km is the mission scope; altitudes outside raise a domain error instead of a number. Figure 5 shows the model against the published USSA 1976 checkpoint values that the test suite pins (5–80 km, temperatures, pressures and densities at 5×10^{-5} relative tolerance); the environment suite of 41 tests additionally checks layer-boundary continuity, monotone decay, exact tail scale height, hydrostatic balance and the domain rejection behavior.

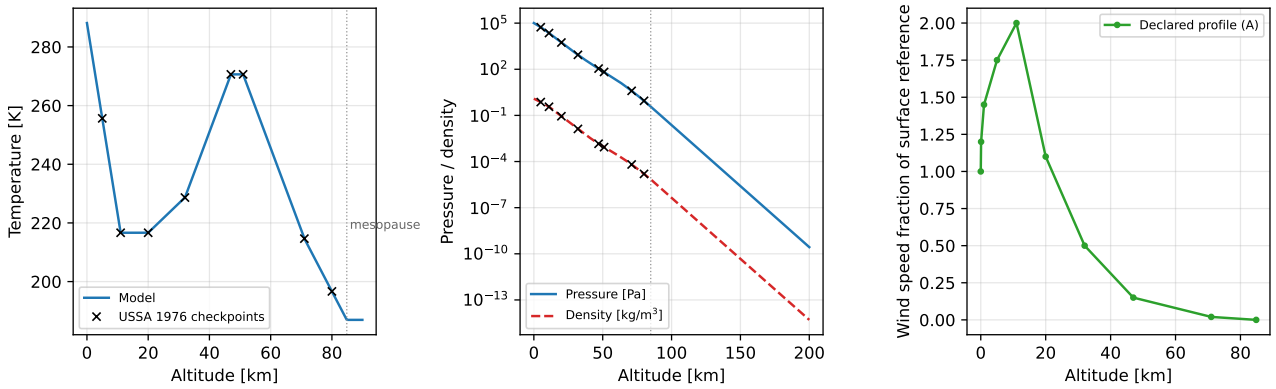


Figure 5 Environment model. Left: layer temperatures to the mesopause with published USSA 1976 checkpoint values. Center: pressure and density (logarithmic) to 200 km, showing the isothermal tail above the mesopause; black markers are the published checkpoints. Right: the declared wind-profile scale factor of the surface reference wind (class A), which is zero above the modeled atmosphere; the nominal mission runs in calm air.

4.2 Gravity and rotating-frame accelerations

Gravity is the reduced Newtonian point-mass model. With the Earth radius $R_E = 6\,371\,\mathrm{km}$ and the configured surface gravity g_0, the gravitational parameter is $\mu = g_0 R_E^2$, so the gravitational acceleration at the launch site is the configured value by construction:

$$\mathbf{a}_g = -\frac{\mu}{\|\mathbf{r}\|^3}\,\mathbf{r}. \tag{15}$$

Gravity never appears as a force; Equation (15) is added once as an acceleration next to thrust over mass. The module also provides the 3×3 gravity-gradient map $\nabla_{\mathbf{r}}\mathbf{a}_g = (\mu/\|\mathbf{r}\|^3)\,(3\,\hat{\mathbf{r}}\hat{\mathbf{r}}^{\mathsf{T}} - \mathbf{I})$, used only in the navigation filter's prediction Jacobian (Section 7), not for torques. Oblateness (J_2) is out of scope.

The external frame is Earth-centred and launch-aligned (the shared conventions are fixed in the architecture chapter of this report); it rotates with Earth relative to inertial space. Translation uses the separate frame acceleration in Equation (16)

$$\mathbf{a}_{\mathrm{fr}} = -2\,\boldsymbol{\omega}_E \times \mathbf{v} \;-\; \boldsymbol{\omega}_E \times (\boldsymbol{\omega}_E \times \mathbf{r}), \tag{16}$$

the Coriolis and centrifugal terms for velocity relative to the rotating Earth, with the rotation vector $\boldsymbol{\omega}_E = \Omega\,(\cos\varphi,\ 0,\ \sin\varphi)$ at launch latitude φ. The nominal mission uses the sidereal rate $\Omega = 7.2921150 \times 10^{-5}$ rad/s at $\varphi = 28.6°$; setting $\Omega = 0$ disables the terms. The frame accelerations, and their analytical gradients, are supplied once per right-hand-side evaluation next to gravity. The angular equations and the gyro model retain the project's approximate Earth-rate treatment: no complete inertial-rate coupling is claimed.

4.3 Wind and ground reference

The wind model is a surface reference vector (ECEF components in the mission configuration) that can be rotated about the launch-site vertical, scaled by a declared altitude profile: the scale factor of the surface reference grows through the boundary layer, peaks at 2.0 around the 11 km jet-stream core, decays through the stratosphere and reaches exactly zero at the mesopause — there is no wind above the modeled atmosphere (Figure 5, right). The profile is a declared stylization (class A). There are no gusts, no time dependence and no direction shear, and the wind enters the physics only through the relative-velocity term of the aerodynamics model. The nominal full-mission run uses calm air: the mission configuration sets no surface reference wind, so the default zero vector applies throughout.

The ground is a bare sphere of radius R_E; altitude is measured radially above it, with an optional reference-level shift, and supports a station offset from the center of gravity (used, for example, for engine-exit altitude). Terrain and contact forces are out of scope; contact is evaluated at the mission level through the leg-tip geometry discussed with the landing criteria in the mission-execution chapter of this report.

Assumptions and limitations The environment is dry air only, with a strictly layered USSA profile; gravity is spherical with no oblateness; the wind is a declared stylization without gusts, time dependence or direction shear; the ground is a bare sphere without terrain; and the frame-acceleration treatment is exact for translation while the attitude chain carries only the approximate Earth-rate treatment noted above. These scope choices (class A) apply uniformly to ascent and return; no environment parameter was fitted to flight data. The module validates its configuration at construction — non-finite gravity, radius, ground offset or wind, non-positive radius, a negative rotation rate or a latitude beyond ±90° all raise an error before flight, not mid-integration.

5 Aerodynamics

`Aerodynamics.py` returns Body-frame forces and CG moments using the evidence-qualified register `data/aero_configurations.json`. Four configurations serve stack/booster phases; a fifth, `second_stack`, is declared but not selected by the independent upper-stage drag provider (Section 8).

5.1 Flow state and force build-up

Equation (17) defines the relative flow and total angle:

$$\mathbf{v}_{\text{rel}} = \mathbf{v}_{\text{CG}} - \dot{\mathbf{c}} - \mathbf{w}, \qquad \alpha = \operatorname{atan2}\big(\|\mathbf{v}_{\text{rel},yz}\|,\ v_{\text{rel},x} \big), \tag{17}$$

where $\dot{\mathbf{c}}$ is the axial center-of-gravity migration rate (Section 3, stage depletion) and $\mathbf{w}$ the wind in Body components (Section 4). The total flow angle α spans 0 (nose-first) to π (engine-first); every direction in the lateral plane is handled, not only a pitch plane. The dynamic pressure is $\bar{q} = \frac{1}{2}\,\rho\, v_{\text{rel}}^2$ with the atmosphere state of Section 4.

The axial and potential-normal loads follow Equation (18); side segments add the crossflow terms of Equation (19):

$$F_x = -\,\bar{q}\, S_{\text{ref}}\, CA_0(M)\ \cos\alpha\,|\cos\alpha|, \qquad N = \bar{q}\, S_{\text{ref}}\, C_{N\alpha}\ \sin\alpha\,|\cos\alpha|, \tag{18}$$

$$D_i = \bar{q}\ C_{Dc,i}\, S_i\ \sin^2\alpha \qquad \text{per side segment } i, \tag{19}$$

Reference areas/lengths appear in Table 7. Normal force opposes lateral flow at a windward-end station (57.6 m for the stack, 41.0 m for the booster), mirrored about CG for retro flow. Crossflow forces act at side-area centroids. The landing configuration adds fin and deployed-leg segments to the body model.

The $\cos\alpha|\cos\alpha|$ factor reverses the axial load and fixed-deflection fin effect in engine-first flight and makes them vanish at 90°. Station moments are assembled once as $(\boldsymbol{s} - \boldsymbol{c}) \times \boldsymbol{F}$. Geometry determines restoring or destabilizing effects; the model does not impose passive stability.

Equation (20) closes the moment set with per-axis damping:

$$\mathbf{M}_{\text{damp}} = \operatorname{diag}(c_l, c_m, c_m)\ \frac{\bar{q}\, S_{\text{ref}}\, L_{\text{ref}}^2}{2\, v_{\text{rel}}}\ \boldsymbol{\omega}, \tag{20}$$

with pitch and yaw coefficient c_m tied by symmetry and roll coefficient c_l separate (declared values: -10 full stack, -20 booster, roll -3 with fins deployed; a zero coefficient means that no roll-damping contribution is applied).

5.2 Configurations and axial curves

Configuration	Phase	S_{ref} [m^2]	L_{ref} [m]	Axial curve / fins
`full_stack`	launch to separation	21.08	62.4	sourced M 0–7; declared M 7–10
`booster_alone`	coast, boostback	10.52	42.6	declared (±50 %)
`booster_return`	entry burn, descent	10.52	42.6	declared; four fins
`booster_landing`	landing burn	10.52	42.6	declared; four fins and legs
`second_stack`	stage-2 body (declared)	10.52	19.8	declared, sleeker estimate

Table 7 The five aerodynamic configurations. Station datum is the configuration's own structural base. Fins deploy after boostback; the legs deploy at landing ignition, which is represented by the segment difference between `booster_return` and `booster_landing`.

Figure 6 shows the adopted $CA_0(M)$ curves. The full-stack curve to Mach 7 is read from Pezent's RasAeroII analysis of a Falcon-9-like stack with fairing [7]; its transonic peak is about 1.16 near Mach 1.06. The extension to Mach 10 and all booster/upper curves are engineering estimates, with declared uncertainties around ±50% for the unsourced curves. The fairing-stack curve is transferred to Dragon geometry as an E/A approximation: reference area and coefficient normalization must be retained together. It is not a CRS-10-specific aerodynamic identification.

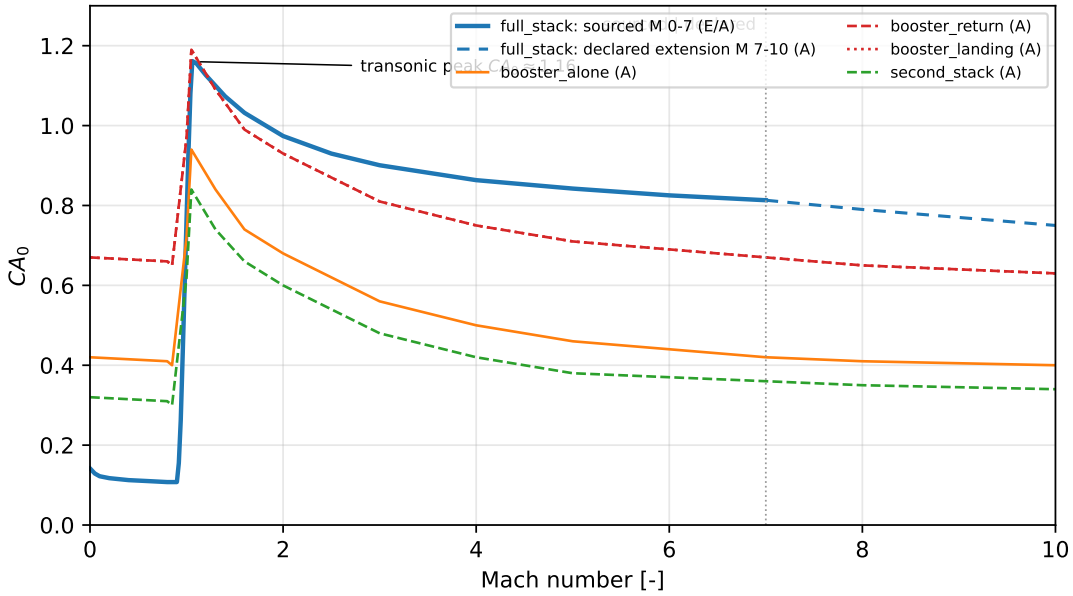


Figure 6 Axial drag coefficient $CA_0(M)$ of the five configurations. Only the full-stack curve to Mach 7 is sourced; everything else is a declared estimate (evidence classes in the legend). The transonic peak of the full stack reaches $CA_0 \approx 1.16$.

5.3 Grid fin control

The return/landing configurations have four fins at 45°, 135°, 225° and 315° about Body $+x$, station 39.6 m and pivot radius 1.83 m. Each uses area 1.65 m^2, effective lift slope 1.2 rad^{-1} and travel ±0.35 rad, all research-model assumptions. Equation (21) gives the force for deflection δ_i and azimuth φ_i:

$$\boldsymbol{F}_i = \bar{q} S_{\text{fin}} c_{\ell,\delta} \delta_i \cos\alpha |\cos\alpha| \begin{bmatrix} 0 & -\sin\varphi_i & \cos\varphi_i \end{bmatrix}^{\mathsf{T}}. \tag{21}$$

The three-dimensional mount arm generates pitch/yaw through axial offset and roll through radial offset. Equal deflections produce roll; differential pairs generate transverse moments. Command selection belongs to Section 8. The aerodynamic interface applies instantaneous deflections without servo dynamics. Mach/angle-dependent effectiveness, plume interference and deflection-induced axial drag are unresolved.

5.4 Domain and verification

The model rejects unsupported Mach values (the curves end at 10), unknown configurations, invalid stations and out-of-travel commands. Declared coefficient uncertainty remains an uncertainty within that domain, not an execution error.

The 42 aerodynamic tests check axial baselines, force direction, retro reversal, moment arms, dissipative damping, continuity, fin combinations and all configuration records, plus a short propagation through the dynamics provider. These checks verify the implementation; they do not validate the unsourced coefficient curves against flight data.

6 Sensor Models and Measurement Generation

Four sensor front-ends generate measurements from the truth trajectory: an inertial measurement unit (IMU), a global navigation satellite system (GNSS) receiver, a light detection and ranging (LiDAR) altimeter, and a star tracker (`sensors.py`). Sampling occurs only at accepted integration times. The nominal sensor periods are integer multiples of the 0.01 s maximum step, so all rates in Table 8 are hit exactly; no random numbers are drawn during integration trial evaluations, and a fixed seed reproduces a run sample for sample. The navigation-consumer side of these streams is the filter of Section 7; this section covers measurement generation only. The numerical specifications are class-A research assumptions, selected to exercise bias, scale, correlated-error and aiding-availability behavior. They are not certified Falcon 9 hardware specifications. Their rates align with the nominal integration clock; independent tests in Section 9.4 assess the estimator under these declared error models.

Sensor	Rate	Mounting	Error model
IMU	100 Hz	station $[38, 0, 0]$	accel $\sigma = 0.01\,\mathrm{m/s^2}$, bias $[0.05, -0.03, 0.02]\,\mathrm{m/s^2}$, scale factor 10^{-3}, bias random walk $10^{-4}\,\mathrm{m\,s^{-5/2}}$; gyro $\sigma = 10^{-4}\,\mathrm{rad/s}$, bias $[0, 0, 10^{-4}]\,\mathrm{rad/s}$, bias random walk $10^{-7}\,\mathrm{rad\,s^{-3/2}}$; misalignment and temperature models supported
GNSS	10 Hz	antenna $[41, 0, 0]$	pos $\sigma = [1.5, 1.5, 2.5]$ m plus Gauss-Markov $\sigma = [0.6, 0.6, 1.0]$ m at $\tau = 300$ s; vel $\sigma = 0.1\,\mathrm{m/s}$; latency 0.1 s; outages and dropouts supported
LiDAR	10 Hz	station $[0.5, 0, 0]$, boresight $-x$	slant range 1–1 800 m, $\sigma = 0.05\,\mathrm{m} + 10^{-4}\,R$
Star tracker	4 Hz	boresight 60° from $+x$ toward $+y$	cross-boresight $\sigma = 5$ arcsec, boresight roll $\sigma = 40$ arcsec; latency 0.1 s

Table 8 Sensor front-ends and adopted settings; station coordinates are in metres in Body axes. Every entry is a declared research assumption (class A), not a certified Falcon 9 sensor.

6.1 Inertial measurement unit

The IMU measures specific force and angular rate at its fixed structural station. The ideal specific force includes the station kinematics of the dynamics core — the angular-rate and center-of-gravity-migration terms of moving the measurement point off the center of gravity — so the accelerometer resolves the modeled structural-point motion. Equation (22) gives the measured pair:

$$\begin{aligned} \mathbf{f}_{\mathrm{meas}} &= \mathbf{A}\,\mathbf{f}_{\mathrm{true}} + \mathbf{b}_a + \mathbf{b}_{a,T} + \mathbf{b}_{a,\mathrm{rw}} + \mathbf{n}_a, \\ \boldsymbol{\omega}_{\mathrm{meas}} &= \mathbf{G}\,\boldsymbol{\omega}_{\mathrm{true}} + \mathbf{b}_g + \mathbf{b}_{g,T} + \mathbf{b}_{g,\mathrm{rw}} + \mathbf{n}_g. \end{aligned} \tag{22}$$

with per-axis scale-factor/misalignment matrices $\mathbf{A}, \mathbf{G} = \mathbf{I} + \mathrm{diag}(\cdot) + \mathbf{M}$ (zero diagonal of $\mathbf{M}$), constant biases, temperature coefficients about a 20 °C reference, bias random walks driven per sample as $\mathcal{N}(0, \sigma_{\mathrm{rw}}^2\,\Delta t)$, and white noise at the sigmas of Table 8. The nominal mission uses the biases, scale factor and random walk listed in the table, with misalignment, temperature and gyro scale-factor terms at zero (modeled but switched off). The gyro measures the Body rate relative to the rotating Earth-centred Earth-fixed frame (ECEF) frame, consistent with the quaternion transport equation the navigation filter propagates. This is an A-class relative-rate convention: a physical inertial gyro would additionally include Earth rate transformed into Body axes. The star-tracker ECI conversion does not supply that missing gyro term.

6.2 GNSS, LiDAR and star tracker

The GNSS receiver provides antenna position and velocity in the external frame, transformed from the truth state with the full lever-arm kinematics (antenna station minus IMU station, angular-rate cross term and center-of-gravity migration). A 0.1 s latency model feeds a truth-side history buffer that is linearly interpolated at the

delayed epoch, so the emitted sample is a measurement of the past state, not of the present one. Before the full delay history exists, the epoch is limited to the oldest available state. The recorded epoch and effective latency identify this startup behavior. The position error is a first-order Gauss-Markov process beside the white noise (updating per sample with the configured 300 s correlation time), modelling the slow, correlated component absent from naive white-noise GNSS error. First-fix delays, scheduled outages and random dropouts are supported and reported through a validity flag; the nominal run has none of them, and every sample is valid.

The LiDAR measures slant range to the spherical ground along its $-x$ boresight. Validity is geometric: the analytic line–sphere intersection must exist, point downward and land inside the declared 1–1 800 m envelope; invalid geometry is reported as an invalid sample, never extrapolated. Range noise is $0.05\,\mathrm{m} + 10^{-4}R$.

The star tracker provides absolute attitude as a sensor-frame-to-Earth-centred inertial frame (ECI) quaternion. The ECI frame is defined to coincide with the launch-aligned ECEF frame at $t = 0$, and the Earth rotation carries the conversion, so the injection of rotation into the attitude chain is explicit rather than implicit. Noise is 5 arcsec cross-boresight and 40 arcsec about the boresight per exposure, with a 0.1 s latency applied on the attitude history. Availability is physical: the unit is blinded above $2^\circ/\mathrm{s}$ body rate, below 60 km altitude (dense atmosphere), inside the 20° Earth-limb exclusion, and optionally inside a 40° solar exclusion; first-fix, outage and dropout faults are supported (nominally off). The reported sample always carries its blinding reason, making the availability record auditable in the run output.

Assumptions and limitations All noise, bias and latency values are declared assumptions, not datasheet values; the GNSS model has no ionospheric, tropospheric or multipath terms; the LiDAR assumes the bare sphere of Section 4 as ground; the star tracker mounting is a declared mounting assumption that keeps the boresight sky-pointing for an upright vehicle, and vehicle-generated obscuration (plumes, stage interfaces) is not modeled.

7 Navigation Architecture and State Estimation

The multiplicative error-state Kalman filter (MEKF) in `kalman.py` estimates motion at the fixed IMU station from the measurements of Section 6. Its quaternion is propagated from measured gyro rates and corrected by observations; true attitude is used for initialization, sensor generation and evaluation, rather than as a continuous navigation input. Vehicle commands use physical-state feedback, so navigation accuracy and landing success are evaluated separately.

7.1 State, calibration and initialization

The nominal state stores position and velocity in frame E, Body accelerometer bias, the Body-to-E quaternion, gyro bias, three gyro scale factors, six gyro cross-axis terms, three accelerometer scale factors and three GNSS position-bias terms. It contains 31 scalars, including a four-component unit quaternion. The covariance has 30 independent coordinates, given by Equation (23):

$$\begin{aligned} \delta\boldsymbol{x} &= [\delta\boldsymbol{p}^\mathsf{T}, \delta\boldsymbol{v}^\mathsf{T}, \delta\boldsymbol{b}_a^\mathsf{T}, \delta\boldsymbol{\theta}^\mathsf{T}, \delta\boldsymbol{b}_g^\mathsf{T}, \delta\boldsymbol{s}_f^\mathsf{T}, \delta\boldsymbol{m}^\mathsf{T}, \delta\boldsymbol{s}_a^\mathsf{T}, \delta\boldsymbol{b}_G^\mathsf{T}]^\mathsf{T}, \\ \boldsymbol{q}_{\text{true}} &= \hat{\boldsymbol{q}} \otimes \operatorname{Exp}(\delta\boldsymbol{\theta}). \end{aligned} \tag{23}$$

Here Exp converts a rotation vector into a unit quaternion, and $\delta\boldsymbol{\theta}$ is a Body-axis error [19]. Gyro calibration uses $\hat{\mathbf{G}} = \mathbf{I} + \operatorname{diag}(\hat{\boldsymbol{s}}_f) + \mathbf{M}(\hat{\boldsymbol{m}})$, where $\mathbf{M}$ holds the six off-diagonal terms; accelerometer calibration uses $\hat{\mathbf{D}}_a = \mathbf{I} + \operatorname{diag}(\hat{\boldsymbol{s}}_a)$. The colored GNSS error $\boldsymbol{b}_G$ is expressed in E.

Table 9 defines the A-class initialization. Position starts with the fixed perturbation $[1, -1, 0.5]^\mathsf{T}$ m; velocity and mean attitude start at the launch state. All estimated bias and calibration means are zero. This is an assumed initial alignment, not a measured alignment solution.

Error block	Size	Units	Initial standard deviation per coordinate
Position / velocity	3 + 3	m / m s^{-1}	2 / 0.05
Accelerometer bias	3	m s^{-2}	0.1
Attitude / gyro bias	3 + 3	rad / rad s^{-1}	0.01 / 10^{-3}
Gyro scale / cross-axis terms	3 + 6	– / rad	10^{-3} / 10^{-3}
Accelerometer scale	3	dimensionless	10^{-3}
GNSS colored position bias	3	m	$[0.6, 0.6, 1.0]^\mathsf{T}$ (stationary prior)

Table 9 Nominal 30-dimensional error covariance. The quaternion has four stored components but three independent error coordinates. Initial cross-covariances are zero.

7.2 IMU and covariance prediction

Consecutive raw specific-force and gyro samples are averaged over each interval and recalibrated using the current parameter estimates. With $\mathcal{N}(\boldsymbol{q}) = \boldsymbol{q}/\|\boldsymbol{q}\|$, Equation (24) defines prediction; $\mathbf{C}_{1/2}$ uses the half-interval quaternion:

$$\begin{aligned} \overline{\boldsymbol{\omega}} &= \hat{\mathbf{G}}^{-1}(\overline{\boldsymbol{\omega}}_{\text{meas}} - \hat{\boldsymbol{b}}_g), & \overline{\boldsymbol{f}}_c &= \hat{\mathbf{D}}_a^{-1}(\overline{\boldsymbol{f}}_{\text{meas}} - \hat{\boldsymbol{b}}_a), \\ \hat{\boldsymbol{q}}_{k+1}^- &= \mathcal{N}[\hat{\boldsymbol{q}}_k^+ \otimes \operatorname{Exp}(\overline{\boldsymbol{\omega}}\Delta t)], & \hat{\boldsymbol{a}}_k &= \mathbf{C}_{1/2}\overline{\boldsymbol{f}}_c + \boldsymbol{g}(\hat{\boldsymbol{p}}_k) + \boldsymbol{a}_{\text{fr}}(\hat{\boldsymbol{p}}_k, \hat{\boldsymbol{v}}_k), \\ \hat{\boldsymbol{p}}_{k+1}^- &= \hat{\boldsymbol{p}}_k^+ + \hat{\boldsymbol{v}}_k^+\Delta t + \tfrac{1}{2}\hat{\boldsymbol{a}}_k\Delta t^2, & \hat{\boldsymbol{v}}_{k+1}^- &= \hat{\boldsymbol{v}}_k^+ + \hat{\boldsymbol{a}}_k\Delta t. \end{aligned} \tag{24}$$

The first IMU sample initializes the averaging/history buffers without advancing time. The covariance transition includes gravity/frame gradients, force-attitude coupling, and gyro bias/calibration effects on both full-interval attitude and midpoint force rotation. For rotation vector $\boldsymbol{\phi}$, the right Jacobian is $\mathbf{J}_r = \mathbf{I} - (1 - \cos\phi)[\boldsymbol{\phi}]_\times/\phi^2 + (\phi - \sin\phi)[\boldsymbol{\phi}]_\times^2/\phi^3$, where $\phi = \|\boldsymbol{\phi}\|$ and a series handles small ϕ [19]. This derivative avoids replacing a finite rotation's sensitivity with the identity.

The discrete prediction is $\mathbf{P}_{k+1}^- = \mathbf{F}_k\mathbf{P}_k^+\mathbf{F}_k^\mathsf{T} + \mathbf{Q}_k$. White sample-noise PSDs are $S_a = \sigma_a^2/f_{\text{IMU}}$ and $S_g = \sigma_g^2/f_{\text{IMU}}$; acceleration-noise blocks scale as $S_a\Delta t^3/3$, $S_a\Delta t^2/2$ and $S_a\Delta t$ for position, position–velocity and velocity, with calibrated force rotation applied. Bias random walks grow covariance proportionally to Δt; calibration means are constant and nominal calibration random walks are zero.

The GNSS colored-bias model matches the generator's first-order Gauss–Markov process: $\hat{\boldsymbol{b}}^{-}_{G,k+1} = \varphi \hat{\boldsymbol{b}}^{+}_{G,k}$, $\varphi = \exp(-\Delta t/\tau)$, and $\mathbf{Q}_G = \mathrm{diag}(\sigma_G^2)(1-\varphi^2)$. Thus correlated error is represented by a state, while GNSS measurement covariance contains the remaining white component. The stationary bias covariance is not counted again as independent white noise at each fix.

7.3 Measurement correction and attitude injection

GNSS lever arms are removed using estimated attitude and calibrated rate. Delayed position/velocity observations are advanced using the current estimated acceleration. Their covariance retains the added position variance $\ell^2 \mathbf{R}_v$ and position–velocity cross term $\ell \mathbf{R}_v$, for measurement age ℓ. This short-delay, constant-acceleration approximation is not a full out-of-sequence state update. LiDAR uses the analytic ray–sphere model, differentiated through position, attitude, boresight and sensor arm; invalid geometry skips correction.

The star quaternion is converted from ECI to E at its actual exposure epoch. Stored gyro samples are integrated from exposure to reception, using current estimated calibration, to advance the attitude observation. Its Jacobian includes bias/calibration sensitivity across that interval and its noise includes the propagated gyro uncertainty. At startup, latency is limited to the truth history actually available; each saved sample records its epoch and effective age.

For innovation $\boldsymbol{\nu} = \boldsymbol{z} - h(\hat{\boldsymbol{x}}^{-})$, the filter forms $\mathbf{S} = \mathbf{H}\mathbf{P}^{-}\mathbf{H}^{\mathsf{T}} + \mathbf{R}$ and $\mathbf{K} = \mathbf{P}^{-}\mathbf{H}^{\mathsf{T}}\mathbf{S}^{-1}$. Equation (25) gives the Joseph covariance update and multiplicative injection:

$$\begin{aligned} \delta\hat{\boldsymbol{x}} &= \mathbf{K}\boldsymbol{\nu}, & \widetilde{\mathbf{P}} &= (\mathbf{I}-\mathbf{K}\mathbf{H})\mathbf{P}^{-}(\mathbf{I}-\mathbf{K}\mathbf{H})^{\mathsf{T}} + \mathbf{K}\mathbf{R}\mathbf{K}^{\mathsf{T}}, \\ \hat{\boldsymbol{q}}^{+} &= \mathcal{N}[\hat{\boldsymbol{q}}^{-} \otimes \mathrm{Exp}(\delta\hat{\boldsymbol{\theta}})], & \mathbf{P}^{+} &= \mathbf{T}_r \widetilde{\mathbf{P}} \mathbf{T}_r^{\mathsf{T}}, \quad (\mathbf{T}_r)_{\theta\theta} = \mathbf{I} - \tfrac{1}{2}[\delta\hat{\boldsymbol{\theta}}]_{\times}. \end{aligned} \tag{25}$$

Other blocks receive additive corrections and identity reset blocks. The reset is a first-order small-angle approximation; the result is symmetrized. Star fixes pass only when NIS $= \boldsymbol{\nu}^{\mathsf{T}}\mathbf{S}^{-1}\boldsymbol{\nu} \le 16.27$, the selected three-degree-of-freedom 99.9% threshold [20]. GNSS and LiDAR NIS values are diagnostics without equivalent rejection gates.

7.4 Interfaces and verification limits

One stage-1 filter continues through separation; the upper child has no navigation instance. Optional reduced-state/supplied-attitude modes and an unaided gyro quaternion support diagnostics. The nominal gyro scale/cross-axis truth is zero, so separate nonzero-error fixtures are necessary to test those states. Section 9.4 reports the final full-flight errors, independent calibration tests, aiding acceptance and covariance evidence. These verify the stated sensor assumptions and fixtures; they do not establish hardware performance or navigation-fed control robustness.

8 Mission Execution and Numerical Implementation

The mission layer connects the models of Chapters 1–7. Commands are A-class project designs: official schedule entries anchor selected events and state-triggered conditions determine return transitions. This chapter defines execution; observed times belong in Section 9.

8.1 Ascent and upper-stage commands

The ascent program tilts the nose away from local vertical. Equation (26) defines its target angle and ideal angular-acceleration command, with $x = \max[(t - 8\,\mathrm{s})/(100\,\mathrm{s}), 0]$ and Body attitude error $\boldsymbol{e}_R$:

$$\theta_{\mathrm{cmd}} = 80^\circ \frac{x^2}{1+x^2}, \qquad \boldsymbol{\alpha}_{\mathrm{cmd}} = k_P \boldsymbol{e}_R - k_d \boldsymbol{\omega}, \qquad k_P = 2\,\mathrm{s}^{-2}, \quad k_d = 2.5\,\mathrm{s}^{-1}. \tag{26}$$

Here $\boldsymbol{e}_R = \mathbf{C}_{BE}^{\mathsf{T}}(\boldsymbol{n} \times \boldsymbol{n}_{\mathrm{cmd}})$, with actual and commanded unit nose directions in E. The implemented direction uses projected launch-east. Although the configuration contains a 45° heading field, the current controller does not use that field in constructing the target; no reconstructed ISS ascent plane is claimed. The pitch timescale and operating point were calibrated against the ascent record used again in Section 9.3.

All nine engines initially run at full throttle. The shutdown envelope of Equation (11) reduces both thrust and flow from 140 to 141 s; zero-flow guards then permit separation at 144 s. The conserved upper child ignites at 152 and cuts off at 545 s: throttle 0.95, one-second ramps, 200 km height target and a 0.05 s maximum step. Height/radial-speed feedback selects attitude. Its drag uses $C_D = 0.3$ on the stage cross-section, not the declared `second_stack` configuration.

8.2 Return commands and fuel management

Table 10 collects the principal return settings. The three-engine set is the central engine and an opposite outer pair (indices 0, 1, 5); landing uses only the central engine. Restart guards require separation, zero flow, engines off, available fuel and assumed propulsion readiness, plus the required descent condition and pointing error below 25°. Propellant settling and ignition health are assumed rather than simulated.

Setting	Project value and use
Planning fuel allocations	Boostback 38 t; entry 16 t; landing 10 t; protected reserve 2 t
Boostback	Throttle 1; 10–60 s planning window; predicted-miss target 500 m
Entry	Descending through 45 km; throttle 0.8; shutdown target 500 m s^{-1}
Landing ignition	Reduced stopping predictor, factor 1.05 and 1.5 km clearance allowance
Landing targets	Contact vertical velocity −0.5 m s^{-1}; lateral gains 0.0576 s^{-2} and 0.48 s^{-1}
Direction/rate commands	Force-solver tilt 20°, terminal 4° below 20 m; desired rate clipped per axis at ±0.22 rad s^{-1}; angular-acceleration command norm capped at 0.12 rad s^{-2}
Ideal-couple authority	2 MN m during burns; 200 kN m in coast

Table 10 A-class return settings from `data/crs10_mission.json`. Fuel allocations guide burn termination; they are not measured CRS-10 reserves. The limiting-tank residual remains separate from the protected reserve.

Boostback points against the reduced-order predicted impact miss. The predictor advances central gravity, frame acceleration, approximate drag and a prospective entry burn in 0.5 s steps; it is refreshed once per second. Steering opposes the predicted horizontal miss with a time-to-impact gain and an upward bias, then normalizes the result to a direction. The first satisfied miss, fuel or duration condition starts shutdown; a 500 kg margin anticipates ramp consumption.

Entry starts on descending altitude and ends on speed or fuel cap. A constrained force solver selects pointing from desired vertical/lateral acceleration and aerodynamic reaction. Landing starts when the 0.1 s stopping predictor leaves no more than the declared clearance allowance.

Landing feedback operates on the lowest leg tip. If h_ℓ and $\dot{h}_\ell$ are its clearance and radial velocity, the commanded contact acceleration is $[-\dot{h}_\ell|\dot{h}_\ell| - v_t^2]/[2\max(h_\ell, 0.005\,\mathrm{m})]$, with target v_t from Table 10. Rotational, moving-CG and ground-curvature corrections convert this to the required CG acceleration. Lateral position and velocity feedback drive the assumed target. Unused boostback/entry allocations may transfer to landing, while the reserve and limiting-tank residual are protected; a 200 kg margin anticipates landing shutdown. Contact geometry and acceptance limits are defined in Section 3.

8.3 Force/moment allocation and actuator representation

The force solver respects the tilt envelope. Throttle ramps are numerical envelopes, not additional validated steady operating points. Fins receive static deflections through Section 5; travel-rate dynamics are absent.

Return attitude control is cascaded: a direction error produces a desired angular rate with gain k_p, then $k_d(\omega_{\text{des}} - \omega)$ produces angular acceleration. Both gains have units s^{-1}: $k_p = 1$, $k_d = 2$ normally, and $k_p = 1.5$, $k_d = 6$ during landing. This differs from the direct ascent PD law in Equation (26). Simultaneous transverse rate commands can produce a larger actual rate magnitude than a per-axis clip. Neither that clip nor the acceleration-command cap guarantees a bound on the vehicle's measured state.

Equation (27) distinguishes the ideal attitude input from physical TVC:

$$\boldsymbol{M}_{\text{ideal}} = \mathbf{J}_C \boldsymbol{\alpha}_{\text{cmd}}, \qquad \mathbf{J}_C^{-1} \boldsymbol{M}_{\text{ideal}} = \boldsymbol{\alpha}_{\text{cmd}}. \tag{27}$$

The ascent couple is unlimited. Return control adds gyroscopic feedforward and aerodynamic-moment compensation, allocates fin demand, then saturates the residual ideal couple at the declared authority. Changing-inertia and exhaust-flow terms still enter the physical dynamics. Inertia cancellation in the commanded contribution is model inversion, not a loss of inertia or proof that aerodynamic stability removes the need for TVC. `Actuators.py` remains empty and the propulsion evaluator rejects nonzero gimbal angles. Thus landing results test an assumed moment authority, not nozzle/servo dynamics or closed-loop robustness to actuator/model error.

8.4 Integration, event and sensor timing

The accepted-state loop follows this order:

1. Evaluate phase conditions and accept a due event; apply engine/configuration changes and preserve both children at separation.
2. Generate due sensor samples and update navigation. Record truth/estimate output when due.
3. Select a step bounded by 0.01 s, scheduled events, control/ramp knots, output time and the next sensor clock. Near contact, clearance and speed impose a tighter bound.
4. Apply RK4 to the physical state with trial-state loads and mass properties, then synchronize the accepted vehicle object.

RHS evaluations are pure with respect to clocks and random sampling. Guidance normally updates every 0.2 s; the landing clock permits intervals between 0.01 and 0.2 s. Sensors retain their independent clocks from Table 8, and output is normally every 0.1 s. The internal translation representation follows Section 3.

When a trial crosses the ground sphere at a leg tip, bisection refines the contact epoch with up to 35 iterations and a 10^{-9} s interval criterion. Acceptance uses the pre-contact state; position, speed or attitude are not reset to create success. Propellant-limit, model-domain and unmet-event failures produce explicit run status. A full run requires both an acceptable contact and completion of the intended return sequence.

8.5 Two-vehicle execution and reproducible operation

The booster is advanced until contact or failure. The saved upper-child state is then propagated on its own time interval, using the same mission clock; the two trajectories are physically independent after separation even though their computations are executed sequentially. Navigation remains attached to stage 1.

From the repository root, install the pinned dependencies and execute:

```
python -m pip install -r requirements.txt
python main.py --case full --no-show
```

The full case requests 600 s; booster output ends at contact and the upper child continues. `--output` selects a fresh directory; `--plot-only` reopens saved results without integration.

Outputs are truth for both vehicles, measurements, navigation, events/timeline, ascent comparison data and `summary.json`. The summary retains settings, seed, status and input hashes. Section 9 identifies the records behind its figures and tables; Appendix A.5 gives explicit run settings, file contracts and audit fingerprints.

9 Verification, Validation, and Simulation Results

This chapter separates verification of implemented equations, comparison with public flight evidence, and navigation error against simulation truth. The final nominal record R1 uses seed 1, 0.01 s maximum step, 100 Hz IMU and 0.1 s output; its original path is given in Appendix A.5. All 31 runtime/data fingerprints match the reviewed source. The shared archive and `report/reproduce.py` regenerate the result panels and numerical macros without another full-flight integration. The provenance ledger `report/figure_sources.json` records source hashes and extracted metrics.

9.1 Model verification and state continuity

The vehicle/navigation suite contains 130 passing assertion-based tests; the environment suite adds 41. They cover mass aggregation and derivatives, force-arm signs, fixed-mass free fall and constant-moment rotation, rotating Body storage, propellant bounds, aerodynamic flow reversal/damping, event guards, separation and two-child execution. Navigation tests add tolerance-based finite differences for the 30-state transition, SO(3) right Jacobian and sensor sensitivities, as well as startup epoch, delayed-attitude, gate and colored-noise checks. The legacy print-only diagnostics are not counted as assertion tests. These checks establish behavior in their fixtures, rather than agreement with real hardware.

At 144 s, parent mass is 211019.374 kg and child masses are 91089.374 and 119930.000 kg. Their mass residual is zero. Relative linear- and angular-momentum residuals are 9.95×10^{-19} and 9.54×10^{-12}; recombined-CG position error is 4.66×10^{-10} m. These satisfy the numerical criteria of Section 3 for the assumed zero-impulse map. Both child states remain finite and are propagated without replacing their inherited initial conditions. Conservation under an assumed map does not establish the accuracy of real separation impulses or discarded hardware.

9.2 Continuous mission and separated trajectories

Figure 7 shows the complete generated sequence, and Figure 8 distinguishes the two children. Shutdown completes at 141 and separation occurs at 144 s. The three return burns and their measured consumption appear in Table 11; their endpoints are integration results of the conditions in Section 8.

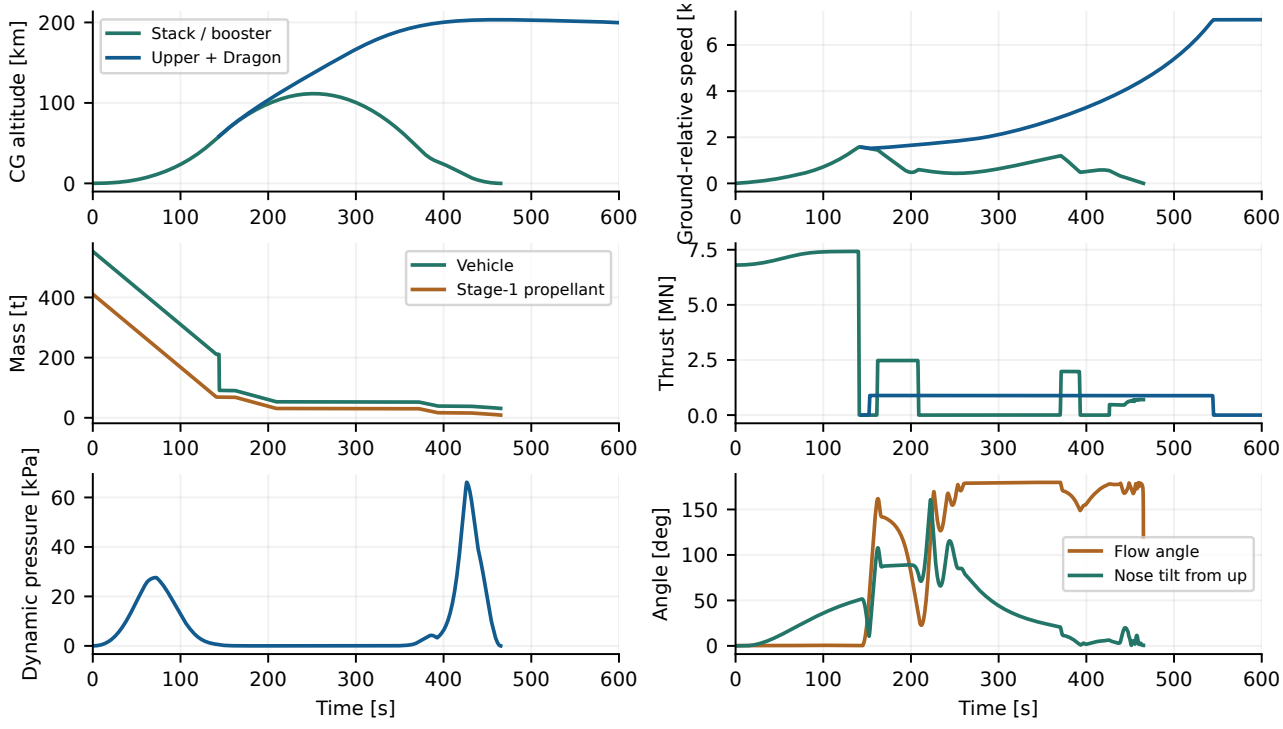


Figure 7 R1 continuous flight. Green denotes stack/booster and blue the upper child where both are shown. The mass jump is a change of tracked configuration at separation. Flow angle near 180° means engine-first relative flow, not attitude-estimation error. The trajectories use project commands and truth-state feedback.

Burn	Start–end [s]	Consumed [t]	End condition
Boostback	161.00–208.81	37.982	Fuel/duration planning cap
Entry	370.14–392.95	14.158	Speed target, then smooth shutdown
Landing	426.16–465.083	7.818	First contact

Table 11 Observed R1 return-burn intervals and fuel consumption. Boostback reaches its planning cap rather than the 500 m predicted-miss target. Its recorded predicted miss at burn end is about 3.97 km; later phases provide additional correction.

The ascent dynamic-pressure maximum is 27.62 kPa at 71.9 s, compared with the approximate 75 s preflight marker. The full-flight maximum is instead 66.16 kPa near landing ignition at 426.3 s; the two maxima must not be confused. Between entry shutdown and landing ignition, gravity accelerates the unpowered descending booster again, explaining the late rise in its speed. Landing then brakes it to the contact state.

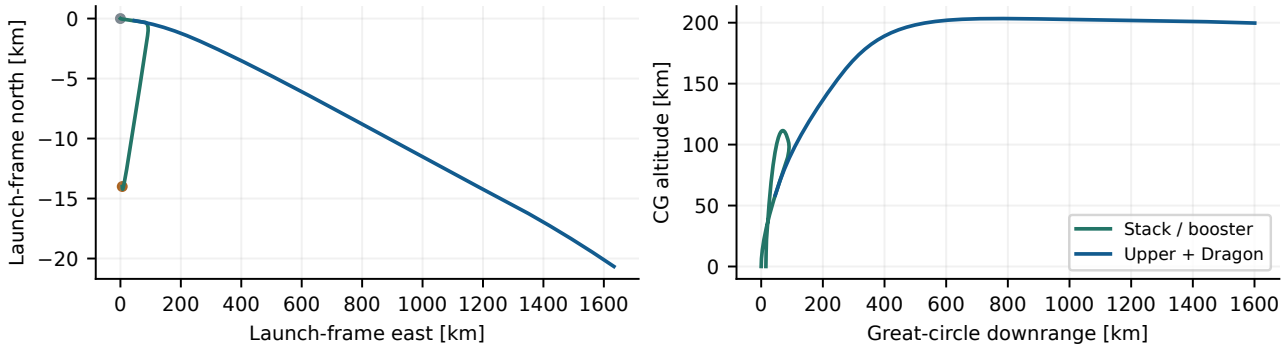


Figure 8 R1 separated paths: launch-frame horizontal projection and altitude against spherical downrange. The marked horizontal points are launch and the assumed return target. The booster returns while the upper-stage/Dragon child continues downrange from the same separation event.

Figure 9 completes the kinematic outputs requested in the assignment [11, slide 5]. Great-circle downrange is $R_E \arccos(\hat{\boldsymbol{r}} \cdot \hat{\boldsymbol{r}}_{\text{pad}})$; it is distance from the pad, not accumulated path length. Flight-path angle is $\gamma = \operatorname{atan2}(v_r, \|\boldsymbol{v} - v_r\hat{\boldsymbol{r}}\|)$, where $v_r = \boldsymbol{v} \cdot \hat{\boldsymbol{r}}$ and $\boldsymbol{v}$ is CG Earth-relative velocity. Actual rate magnitude peaks at $17.825^\circ\,\text{s}^{-1}$ at 223.7 s in return coast. This is compatible with the per-axis desired-rate clip of Section 8; it is not a state-norm limit.

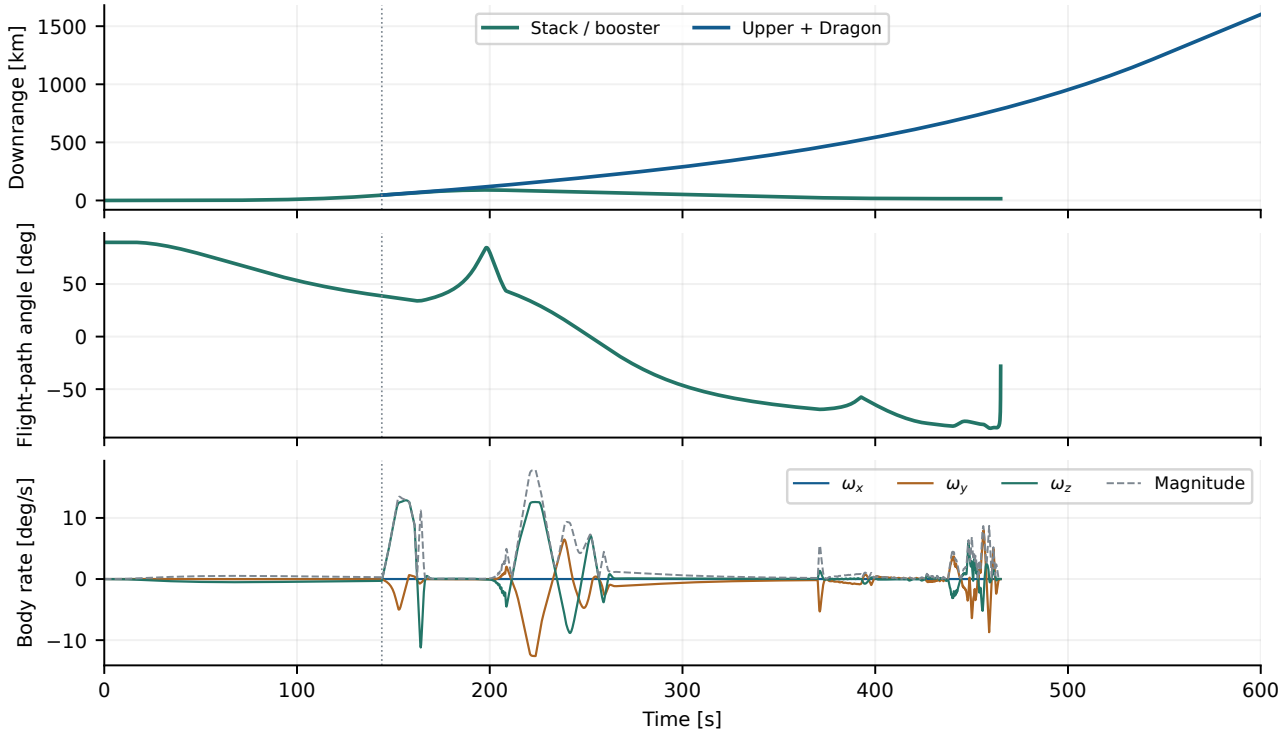


Figure 9 R1 downrange for both children, stack/booster flight-path angle, and true Body-rate components/magnitude. Positive flight-path angle denotes outward motion; negative denotes descent. The angle is omitted below 1 m s^{-1}, where direction is poorly defined. The dotted line marks separation. The post-boostback coast turn explains the largest rate excursion.

Table 12 evaluates the project limits at contact. Contact occurs at 465.0827 s with 8.931 t of stage-1 propellant remaining. The ground condition applies to the lowest leg tip, so the CG remains above zero altitude. The upper child continues to 600 s, reaching 199.694 km and 7.097 km s^{-1}, with 5.472 t of propellant. This is a successful project scenario, not proof of SpaceX guidance reconstruction or validated orbital insertion.

Acceptance metric	R1 result	Project limit
Vertical speed [m s^{-1}]	0.4975	2
Horizontal speed [m s^{-1}]	0.9016	1
Tilt [deg]	0.5349	5
Angular-rate magnitude [deg s^{-1}]	0.01289	1
Horizontal target error [m]	3.306	10

Table 12 R1 mission-level contact metrics. Speeds and target error use the structural-base state; ground contact is tested at the lowest leg tip. Ground contact and deployed legs are also required and satisfied. These are assumed project limits; horizontal speed has the smallest fractional margin.

9.3 Comparison with the ascent reference

Figure 10 compares 140 retained video-derived samples over 0–139.973 s [13, 14]. Altitude RMSE is 1.577 km and speed RMSE 18.46 m s^{-1}. At the last sample, simulated altitude is 54.628 km versus 59.400 km, a −8.03% difference; speed is 1574.19 versus 1609.44 m s^{-1}, a −2.19% difference.

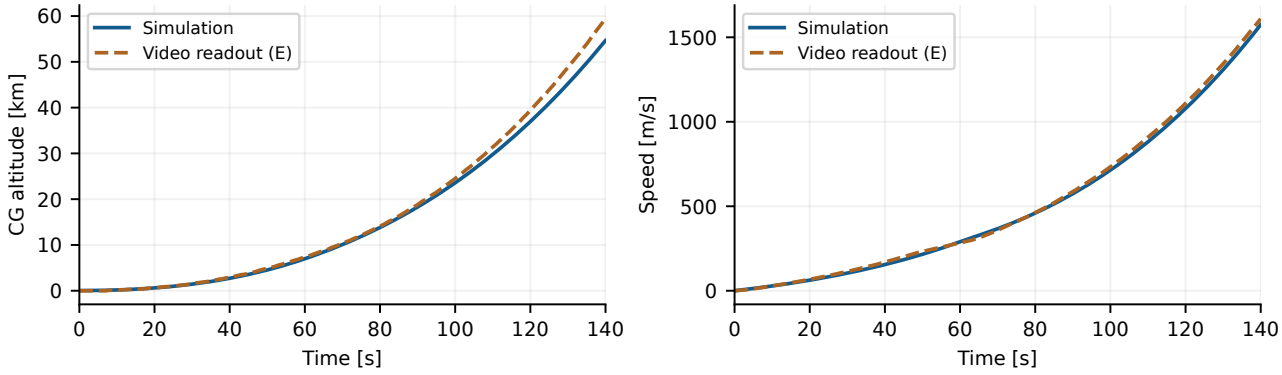


Figure 10 R1 ascent against the E-class digitized broadcast readouts. This is the same record used to calibrate the ascent operating point/pitch timescale. Model CG altitude and ground-relative speed are compared with readouts whose detailed datum/frame conventions are not independently established.

The comparison supports quantitative calibration within the retained ascent interval. It does not independently validate the tuned parameters, establish internal mass properties, or validate the return/upper-stage commands. Readout rounding, digitization and speed-frame uncertainty also limit interpretation of the residuals as pure physical-model error. The fairing-to-Dragon aerodynamic transfer and mixed-source vehicle data remain model uncertainties.

9.4 Navigation accuracy and statistical diagnostics

The gyro-driven navigation result is evaluated at the fixed IMU station, not at the moving CG. Equation (28) defines the three-dimensional position RMS and shortest quaternion-error angle used below:

$$\mathrm{RMS}_p = \sqrt{\frac{1}{N}\sum_{k=1}^{N} \|\hat{\boldsymbol{p}}_k - \boldsymbol{p}_{\mathrm{true},k}\|^2}, \qquad e_q = 2\arccos\left(\left|\hat{\boldsymbol{q}}^{\mathsf{T}}\boldsymbol{q}_{\mathrm{true}}\right|\right). \tag{28}$$

The quaternion inner product is clamped numerically to $[0, 1]$. Velocity RMS uses the corresponding velocity error. Figure 11 shows accuracy and uncertainty diagnostics together.

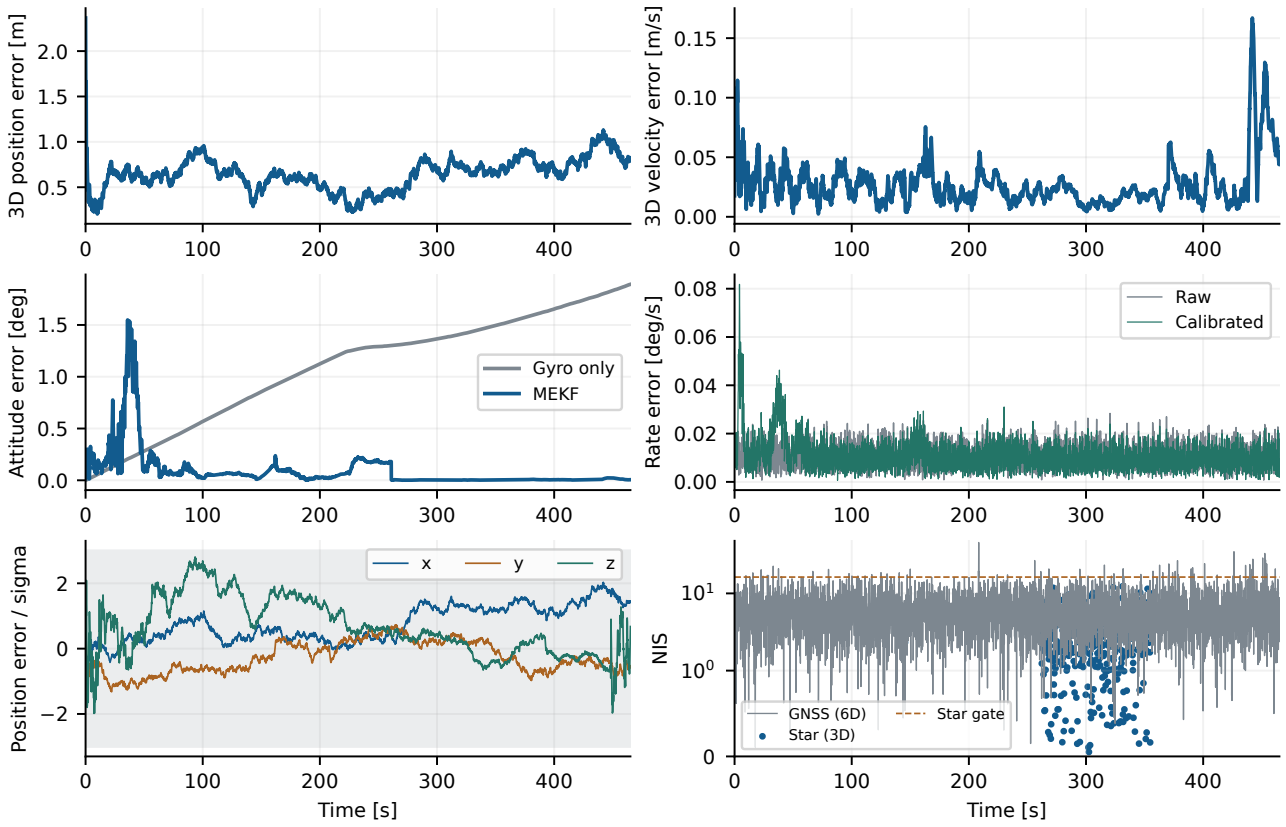


Figure 11 R1 navigation at the IMU station: position/velocity errors, estimated versus unaided attitude error, raw/calibrated angular-rate errors, normalized position errors and innovation diagnostics. The shaded interval is ±3 reported standard deviations. The star gate is unchanged at 16.27; GNSS NIS is diagnostic. Accuracy, covariance consistency and rate calibration are assessed separately.

Position RMS is 0.671 m, velocity RMS 0.0338 m s^{-1}, and attitude RMS 0.208°. Attitude error peaks at 1.549° near 35.9 s and ends at 0.00566°, versus 1.892° for the paired unaided gyro quaternion. These are simulation-truth errors under the declared alignment and sensor settings.

The archived earlier filter supplies a controlled comparison: both truth trajectories are byte-identical to R1 and all common raw sensor fields are identical sample by sample. Its position/velocity/attitude RMS values are 0.881 m, 0.1354 m s^{-1} and 0.456°. The improvement measures the combined estimator changes of Section 7; it does not isolate each change's contribution. Physical flight and guidance are unchanged.

LiDAR supplies 496 valid corrections. All 381 valid star exposures are accepted, with 0 rejected; the gate remains 16.27. Their mean and maximum NIS are 3.09 and 12.31, and mean six-component GNSS NIS is 6.13. Position errors remain inside reported $\pm 3\sigma$ at every recorded epoch on all three axes. These correlated time samples are diagnostics, not independent coverage probabilities. All 4652 stored 30×30 covariance matrices are finite and symmetric, with minimum eigenvalue 1.01×10^{-12}; full matrices are archived for inspection.

Across 4651 synchronized IMU/output epochs, calibrated angular-rate RMS is 0.01251° s^{-1}, compared with raw 0.01152° s^{-1}. The nominal true gyro scale/cross-axis errors are zero; estimating them adds finite-sample uncertainty and does not guarantee improved instantaneous rates. The output remains an estimate of angular velocity, but this nominal case alone is insufficient evidence of calibration benefit. The independent nonzero-error test below supplies that evidence within its own excitation conditions.

9.5 Independent navigation and covariance tests

The executable `report/validate_navigation.py` supplies controlled truths independently of the CRS-10 trajectory. It runs a three-axis accelerometer-scale fixture, an analytic three-axis attitude fixture with nonzero gyro calibration errors, and 20 independent position/covariance realizations. Tolerances and fixture settings are explicit in Table 13 and Appendix A.5; all ten stored checks, including the isolated RK4 test below, pass. Figure 12 shows convergence and the independent final-epoch errors.

Independent check	Measured result	Acceptance criterion
Accelerometer scale / bias error norm	3.02×10^{-5} / 3.79×10^{-5} m s^{-2}	$< 4 \times 10^{-4}$ / < 0.002 m s^{-2}
Final attitude / gyro bias error norm	$0.00363°$ / 1.21×10^{-6} rad s^{-1}	$< 0.03°$ / $< 2 \times 10^{-5}$ rad s^{-1}
Gyro scale / cross-axis error norm	1.88×10^{-4} / 8.06×10^{-5} rad	$< 4 \times 10^{-4}$ / $< 8 \times 10^{-4}$ rad
Late-window raw / calibrated rate RMS	$0.03751/0.01004°$ s^{-1}	Calibrated RMS smaller
Position ANEES, 20 independent trials	2.377 (3 coordinates)	95% interval $[2.024, 4.165]$
GNSS mean NIS in the same fixture	5.957 (6 components)	Engineering check 5.5–6.5

Table 13 Independent navigation checks. Error norms compare estimated parameters with injected truth. Numerical calibration tolerances and the NIS band are chosen engineering criteria; the ANEES interval has a stated chi-square statistical interpretation.

For $M = 20$ independent final position errors, ANEES $= M^{-1} \sum_i \boldsymbol{e}_i^{\mathsf{T}} \mathbf{P}_{p,i}^{-1} \boldsymbol{e}_i$. The nominal 95% acceptance interval is $[\chi^2_{60,0.025}, \chi^2_{60,0.975}]/20$ [20]. A single final epoch per independently sampled prior/noise realization avoids treating one correlated time series as an ensemble. The result is compatible with this position-marginal test, not proof of whole-flight 30-state consistency. The gyro fixture accepts 1799 and rejects 2 of 1801 valid star exposures at the same gate; no requirement for zero stochastic rejections is imposed.

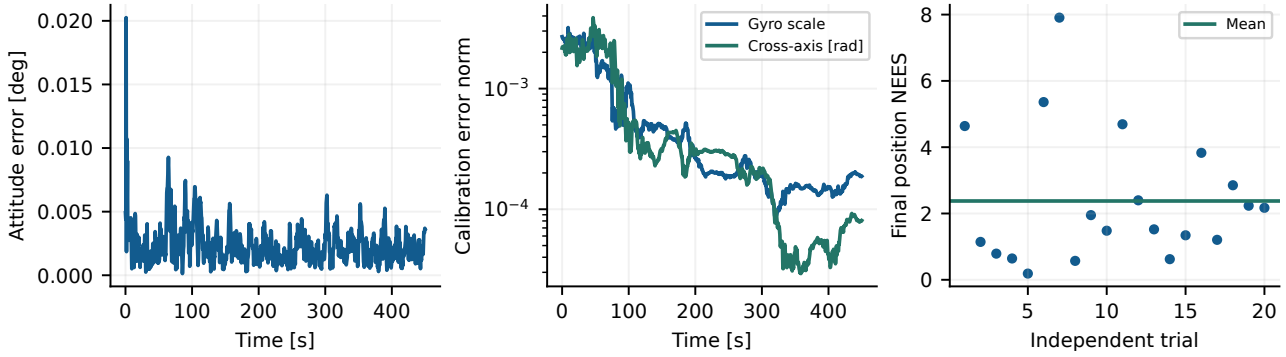


Figure 12 Independent navigation fixtures. Left and middle: bounded analytic three-axis attitude with nonzero gyro bias, scale and cross-axis errors; calibration errors converge under the specified excitation and aiding. Right: one final position NEES per independent realization, with their mean. This is separate from the nominal mission and is not a wind/landing robustness ensemble.

9.6 Numerical sensitivity and demonstrated scope

A saved variant, `results/two-stage-final-check`, uses 0.05 s steps and 20 Hz IMU with matching compared core-physics input hashes. It also lands, at 465.1900 s, with target error 6.171 m and tilt 1.806°, versus R1's 3.306 m and 0.535°. The plan differs in IMU rate and the integration step is also changed. This is a joint step/clock diagnostic, not an isolated RK4 convergence-order study; contact metrics remain numerically sensitive even when both cases pass.

An isolated constant-spin rigid-body fixture tests RK4 without sensor, guidance or event clocks. Against the exact quaternion at 10 s, steps 0.4, 0.2 and 0.1 s give attitude errors 5.1093×10^{-7}, 3.1983×10^{-8} and 1.9997×10^{-9} rad. Step-halving orders are 3.9977 and 3.9994, within the selected $[3.8, 4.2]$ criterion. This verifies fourth-order behavior in the smooth rotation fixture; it does not establish mission-wide convergence through discrete events.

A separate static allocation check quantifies the dry-CG uncertainty of Section 3. With an assumed 10 t of propellant remaining, the booster mass is 32.2 t and nominal CG station 16.211 m. Moving the unallocated 6450 kg centroid over the declared 4.0–40.8 m interval shifts the combined CG by up to ±3.686 m. For an explicitly assumed 1 kN transverse force at a fixed structural station, this changes its CG moment by up to ±3.686 kN m. These are conditional allocation effects, not flight-error confidence bounds.

10 Engineering Assessment and Implications for Future GNC

10.1 What the demonstrated results support

The conservation audit, ascent comparison and independent navigation tests support controlled studies of vehicle physics and navigation. The successful contact in Table 12 uses physical-state feedback and ideal moments; navigation is evaluated separately. Accepted star aiding and nonzero-error calibration are demonstrated, while alignment remains an assumed prior. These distinctions define the simulator's role as the Project 2 foundation for later guidance and control [11, slides 2–4].

10.2 Most dangerous modeling and navigation errors

For the assignment's final question [11, slide 8], the most dangerous errors are incorrect thrust or measurement directions caused by frame or attitude mistakes, and demands beyond physical actuator authority. This assessment concerns consequences; no failure probabilities are estimated. Table 14 connects the risk pathways to the evidence.

Error pathway	Evidence or present scope	Consequence for future GNC
Attitude, rate or frame mismatch	Body-to-E convention is shared; inertial-gyro Earth rate is approximated. Nominal calibrated rate RMS is $0.01251^\circ\,\mathrm{s}^{-1}$, versus raw $0.01152^\circ\,\mathrm{s}^{-1}$.	Thrust, specific force and lever-arm corrections can point in the wrong direction; attitude accuracy alone does not certify rate feedback.
Unavailable control authority	Section 8 uses an unlimited ascent couple and bounded ideal return moments, without nozzle/servo dynamics.	Guidance may request impossible turns; control gains can conceal lag, saturation and loss of authority.
CG, inertia, flow or thrust error	The conditional 10 t-propellant allocation check shifts CG by up to 3.686 m; the engine and internal geometry are mixed-source estimates.	Force arms, angular acceleration, stopping distance and usable fuel change together, potentially invalidating ignition and reserve predictions.
Misleading uncertainty or unavailable aiding	All 381 valid star fixes are accepted in R1; the independent position ANEES test passes. Full-flight ensemble consistency and outage robustness are untested.	Altitude or vertical-speed bias can mis-time braking; untested acquisition or dropout behavior can hide risk despite small nominal RMS error.
Aero, wind, event or timing mismatch	Return coefficients are assumed, R1 is calm, and the joint step/IMU-rate variant changes contact metrics.	Lateral miss, loads and burn timing can exceed nominal margins even when the saved nominal run passes.

Table 14 Consequence-based GNC risk assessment grounded in Sections 3–9. Proposed consequences are engineering reasoning; no probabilities or unperformed robustness results are assigned.

10.3 Recommended progression to guidance and control design

The first recommendation is to extend the demonstrated estimator tests before connecting navigation to control. The present tolerance-based Jacobian checks, nonzero-error fixtures, full covariance logs and independent position ANEES study establish specific behavior; they do not establish full-flight consistency. Further tests should vary initial alignment, aiding outages, delay and nonlinear excitation, including the limits of the first-order attitude reset and short-delay GNSS approximation. Relative-to-E gyro data should be replaced by a physical inertial-rate model with explicit Earth-rate compensation when transferring to hardware.

The second recommendation is to replace ideal moment authority with a physical allocation and actuator model before tuning vehicle control. The model should resolve available engine/gimbal and fin authority, response rate and saturation, and retain the independently calculated CG/inertia rather than treating inversion of inertia as evidence of stability. The $0.902\,\mathrm{m\,s}^{-1}$ nominal horizontal contact speed against the $1\,\mathrm{m\,s}^{-1}$ project limit leaves little demonstrated margin for these omitted effects.

Finally, the isolated RK4 order test should be followed by a full-mission step study with fixed sensor/guidance clocks; sensor-rate studies should hold physical integration settings fixed. Mass allocation, thrust, return coefficients and nonzero wind should be varied before multi-seed navigation-fed trials. Independent flight evidence should be reserved for validation after calibration. These recommended extensions are not completed evidence; Appendix A identifies the present controlled inputs and test settings.

11 Conclusions

Project 2 establishes a physics-driven Falcon 9/Dragon simulator and a sensor-driven navigation system for the selected CRS-10 scenario. Forces and moments generate motion through six-degree-of-freedom equations; changing propellant inventories determine mass, CG and inertia; configuration events preserve inherited child states. The simulator therefore generates the trajectory from its models and commands rather than replaying a prescribed flight record. The shared conventions, module interfaces and reproducible execution provide the computational foundation required for subsequent guidance and control development [11, slides 2–4].

The demonstrated mission connects ascent, shutdown, separation, booster return and contact, while independently continuing stage 2 and Dragon to 600 s. The model/navigation and environment suites total 171 passing assertion-based tests. The separation audit satisfies the selected conservation criteria, and the nominal booster meets all declared contact limits. The isolated quaternion fixture exhibits fourth-order RK4 convergence. These checks verify the implemented scenario and interfaces, rather than certify real Falcon 9 separation or landing performance.

The public ascent comparison quantifies the available external evidence: altitude RMSE is 1.577 km and speed RMSE is 18.46 $\mathrm{m\,s^{-1}}$ over the retained interval. Because this record also informs the ascent calibration, it is not an independent validation set. The return sequence, upper-stage commands and internal mass distribution remain evidence-qualified research models rather than reconstructed CRS-10 flight software or measured mission internals.

Navigation propagates noisy gyro measurements and its own attitude, aided by GNSS, valid LiDAR and 381 accepted star fixes. The nominal IMU-station errors are 0.671 m position RMS, 0.0338 $\mathrm{m\,s^{-1}}$ velocity RMS and 0.208° attitude RMS. Final attitude error is 0.00566°, compared with 1.892° for the paired unaided quaternion. Independent nonzero-error fixtures demonstrate accelerometer/gyro calibration within stated tolerances, and 20 independent realizations pass the position-marginal ANEES test. Nominal calibrated instantaneous rate RMS remains slightly above raw RMS; calibration benefit is demonstrated in the separate nonzero-error fixture. Accuracy, consistency, aiding and parameter observability therefore remain distinct claims.

The principal engineering outcome is a traceable environment in which those distinctions can be investigated. Mixed-source mass properties, assumed return aerodynamics, calm nominal wind, ideal moments and truth-state command feedback limit extrapolation to integrated GNC. Physical actuator modeling, broader uncertainty/outage trials and independent flight validation are justified next steps before closing the navigation-to-control loop. Within this boundary, the project delivers the vehicle, environment, measurement, estimation and numerical interfaces, quantified tests and an auditable assumption/source register. The archived raw evidence and executable figure/validation scripts make the reported results reproducible from a fresh checkout.

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[20] NIST/SEMATECH, *e-Handbook of Statistical Methods: Critical Values of the Chi-Square Distribution*, Section 1.3.6.7.4. https://www.itl.nist.gov/div898/handbook/eda/section3/eda3674.htm

A Reproducibility and Supporting Data

This appendix is the reference register for the active nominal scenario, not a second model derivation. K identifies authoritative published information, E a literature/readout estimate, and A a project assumption. E/A combines sourced information with a geometric, configuration or operating assumption. Computed outputs inherit the evidence limits of their inputs. SI units apply unless explicitly shown otherwise; angles in code are radians, while some input records and plots use degrees.

A.1 Evidence and vehicle-input register

Table 15 resolves the source chain used by the register. The numeric vehicle definitions are in `vehicle_parameters.py`; `data/vehicle_parameters.json` supplies metadata rather than an independent set of numeric overrides. Cargo metadata resolves to the official CRS-10 overview. The executed rotating-frame, two-child and gyro-driven navigation paths are identified in the configuration and saved run; numeric inputs and explicit implementation choices take precedence over older prose descriptions.

Table 15 Evidence sources and applicability.

Input family	Source / locator	Status	Scope limitation
Official mission schedule	[1, PDF p. 2]	K publication; A commands	Approximate preflight times; not measured thresholds or restart states.
Cargo inventory	[2]	K	1530 + 960 = 2490 kg; already inside loaded Dragon mass.
External vehicle geometry	[5, Table 2-1]	K / E/A use	Later guide; exterior diameter, not 2017 internal dimensions.
Legacy stage lengths	[17]	E/A	FT fact sheet plus assumed contiguous stage datums.
Mass/engine research set	[7, Tables 2.16–2.18]	E	Printed p.77 / PDF p.94; not a mission-certified operating set.
Effective sea-level point	[6, Sec.2.2, p.11]	E/A use	Historical 756 kN reference combined with the adopted nozzle model.
Structural tank envelope	[15, Table 1]	E/A	Reverse-engineered structural dimensions, mapped to flat liquid columns.
Dry groups / consumed O/F	[8, Table 2 / Sec.2.2]	E/A / E	Generic research vehicle; group shapes and residual placement assumed.
Initial tank inventories	[9, Table 1, p.215]	E	Only published capacities adopted; no in-flight inventory measurement.
Liquid reference properties	[16, Table 5, PDF p.24]	E	Comparative reference temperatures/densities; pressure unspecified.
Loaded Dragon mass	[3]	E, provisional	External approximate 8430 kg estimate; original host may be inaccessible; no official mass confirmation.
Dragon envelope	[4, p.26 / PDF p.27]	E/A	Earlier Dragon 1 capsule/trunk dimensions, transferred to a uniform equivalent.
Ascent readout	[13, 14]	E	Author-published digitization of broadcast readouts; calibration and comparison share the record.
Atmospheric layers	[10]	K model / A use	Published standard profile, not measured CRS-10 weather.

Table 16 consolidates the independent vehicle inputs and selected derived checks. All stations are from the original stack base along $+x_B$ unless stated as local. No fairing mass or extra cargo mass is added. Engine masses close the stage dry budgets; they are not added a second time.

Table 16 Vehicle and engine input register; paired entries follow the order named.

Parameter	Adopted value	Class	Basis / implementation meaning
Outer diameter / radius	3.66/1.83 m	K	[5, Table 2-1]; radius derived.
Stage-1 / stage-2 lengths	42.6/12.6 m	E/A	[17]; assumed attachment boundary.
Stage-1 dry / initial propellant	22200/410900 kg	E	[7, Table 2.17].
Stage-2 dry / initial propellant	4000/107500 kg	E	[7, Table 2.17].
Loaded Dragon / included cargo	8430/2490 kg	E / K	[3, 2]; cargo is a subset.
Dragon length / radius	7.2/1.83 m	E/A	[4]; 4.4 + 2.8 m envelope, uniform solid cylinder.
Stage-2 local / stack CG	6.3/48.9 m	A	Equivalent uniform midpoint plus 42.6 m offset.
Dragon stack CG	58.8 m	A	42.6 + 12.6 + 7.2/2; not measured.
Total mass / model height	553030 kg / 62.4 m	E/A	Sums of the adopted masses and axial envelopes.
RP-1 / LOX loading	123500/287400 kg	E	[9, Table 1].
Loaded / consumed O/F	2.3271255/2.35	E	Inventory ratio / [8, Sec.2.2].
RP-1 / LOX skin thickness	0.0031/0.0029 m	E	[15, Table 1]; modeled skins.
RP-1 / LOX inner radius	1.8269/1.8271 m	E/A	Outer radius minus the adopted skin; no hardware displacement.
RP-1 / LOX bottom station	4.0/18.7 m	E/A	[15, Table 1]; aligned flat bottoms.
RP-1 / LOX tank length	14.7/22.1 m	E/A	[15, Table 1]; structural envelopes.
RP-1 density / temperature	820 kg m^{-3} / 266 K	E	[16, Table 5]; constant reference property.
LOX density / temperature	1250 kg m^{-3} / 66 K	E	[16, Table 5]; pressure unspecified.
Dry shell: mass / CG / length	8000 kg / 21/42 m	E/A	[8, Table 2]; thin-shell equivalent.
Dry fins: mass / CG	1520 kg / 40.3 m	E/A	[8, Table 2]; equivalent disk.
Dry engines: mass / CG	4230 kg / 0.452 m	E/A	[8, Table 2]; equivalent disk.
Folded legs: mass / CG	2000 kg / 6.66 m	E/A	[8, Table 2]; fixed mass placement.
Residual dry mass / interval	6450 kg / [4, 40.8] m	A	Budget closure; uniform thin-shell equivalent, centroid 22.4 m.
Nominal dry CG	17.52108 m	E/A	Mass-weighted adopted dry allocation.
Dry inertia, axial / transverse	61368.59 / 4441994.04 kg m^2	E/A	Computed component equivalents; not measured inertia.
Stage-1 / stage-2 engine count	9/1	K	Falcon configuration [6]; modeled as central + ring / MVac.
Vacuum thrust, stage 1 / 2	824901/935000 N	E/A; E	756000 + 0.68(101325) / [7, Table 2.16].
Vacuum I_{sp}, stage 1 / 2	311/348 s	E	[7, Table 2.16].
Exit area, stage 1 / 2	0.68/4.90 m^2	E	[7, Table 2.16].
Minimum steady throttle, 1 / 2	0.60/0.40	E	[7, Table 2.16]; off is separate.
Reference pressure / g_0	101325 Pa / 9.80665 m s^{-2}	K	Standard reference constants [10].
Stage-1 sea-level thrust / I_{sp}	756000 N / 285.023 s	E/A	Derived at the reference pressure and retained flow.
Engine dry proxy, stage 1 / 2	470/700 kg	E	[8, Table 2]; inside stage budgets.
Engine ring / pivot, stage 1	1.3/1.0 m	A	Ring radius / axial force pivot; not engine CG.
Stage-2 local pivot	0.5 m	A	From its own base.
Stored gimbal limits, stages 1/2	5°/5°	A	Stored input only; physical gimbal dynamics are inactive.

The stage-1 liquid model is `dual_tank`; stage 2 uses an equivalent uniform propellant cylinder. Inventory reconstruction assumes fixed O/F, no venting/refilling, fixed initial loading, constant density, and settled flat columns. Dry and payload shape equivalents, no slosh, balanced fixed exhaust ports and unchanged folded-leg

mass placement are A-class closures. The 1202.128 kg limiting RP-1 residual is a consequence of the adopted loading/consumption pair, not the separately protected 2000 kg mission reserve.

A.2 Environment and aerodynamic register

Table 17 lists environmental constants and all atmosphere/wind knots. Layer bases use the code's radial-altitude coordinate; the standard profile is a reference atmosphere rather than measured launch weather.

Table 17 Active environment inputs and optional wind-profile definition.

Parameter	Value / knots	Class	Basis / limitation
Earth / ground sphere	$R_E = 6371000\,\mathrm{m}$; ground offset 0 m	A	Spherical Earth approximation; `Environement.py`.
Gravity normalization	$g_0 = 9.80665\,\mathrm{m\,s^{-2}}$; $\mu = g_0 R_E^2$	A	Central field normalized to the chosen surface value.
Earth rate / launch latitude	$7.292115 \times 10^{-5}\,\mathrm{rad\,s^{-1}}$ / 28.6°	K / A	Sidereal-rate constant and rounded scenario latitude; launch-aligned axes.
Air gas constant / heat ratio	$287.05287\,\mathrm{J\,kg^{-1}\,K^{-1}}$ / 1.4	K	USSA reference constants [10].
Sea-level temperature / pressure	288.15 K / 101325 Pa	K	USSA standard model [10].
Layer bases: (h_b, T_b, L)	(0, 288.15, −0.0065); (11000, 216.65, 0); (20000, 216.65, 0.001)	K	h_b [m], T_b [K], L [$\mathrm{K\,m^{-1}}$]; [10].
Remaining layer bases	(32000, 228.65, 0.0028); (47000, 270.65, 0); (51000, 270.65, −0.0028); (71000, 214.65, −0.002)	K	Same units and source.
Mesopause / model ceiling	84852/2000000 m	K / A	Standard-layer endpoint / assumed numerical-vacuum-tail envelope.
Nominal surface wind / rotation	$[0, 0, 0]^\mathsf{T}\,\mathrm{m\,s^{-1}}$ / 0 rad	A	Default calm case; the entire scaled profile remains zero.
Wind knots $(h, extscale)$	(0, 1); (100, 1.2); (1000, 1.45); (5000, 1.75); (11000, 2)	A	Linear interpolation; altitude in m; stylized surface-reference multiplier.
Remaining wind knots	(20000, 1.1); (32000, 0.5); (47000, 0.15); (71000, 0.02); (84852, 0)	A	Exactly zero at and above 84852 m; no gusts/directional shear.

Table 18 gives every declared configuration. Four serve the stack/booster sequence; the declared `second_stack` is inactive in the upper-stage continuation, which instead uses $C_D = 0.3$. $C_{N\alpha}$ is per radian; damping coefficients are dimensionless. Geometry is an equivalent envelope, and all normal-force, damping, fin and booster coefficients are A-class estimates. Only the original full-stack axial readout is E-class; applying its fairing geometry to Dragon remains A-class transfer.

Table 18 Aerodynamic geometry, derivatives and projected side segments from the configuration JSON.

Configuration	$S_{\mathrm{ref}}/L_{\mathrm{ref}}/x_p$ [$\mathrm{m^2/m/m}$]	$C_{N\alpha}/c_m/c_l$	Segments: (bottom m, top m, width m, crossflow coefficient)
`full_stack`	21.08/62.4/57.6	2/−10/0	(0, 55.2, 3.66, 1.1); (55.2, 62.4, 3.7, 1.1)
`booster_alone`	10.52/42.6/41	2/−20/0	(0, 42.6, 3.66, 1.1)
`booster_return`	10.52/42.6/41	2/−20/−3	(0, 42.6, 3.66, 1.1); (38.6, 40.6, 2.4, 1)
`booster_landing`	10.52/42.6/41	2/−20/−3	(0, 42.6, 3.66, 1.1); (38.6, 40.6, 2.4, 1); (0, 3.6, 1.6, 1.2)
`second_stack`	10.52/19.8/17.3	2/−10/0	(0, 12.6, 3.66, 1.1); (12.6, 19.8, 3.7, 1.1)

The common grid-fin control definition in `booster_return` and `booster_landing` has four fins at azimuths 45°, 135°, 225°, 315° (stored rounded radian values), 1.65 m^2 area per fin, slope 1.2 rad^{-1}, pivot radius 1.83 m, station 39.6 m and maximum deflection 0.35 rad. These are A-class estimates, not measured fin effectiveness or actuator limits. The return and landing axial curves are identical; the deployed-leg side segment distinguishes their passive side-force geometry.

Table 19 preserves the complete axial coefficient knots without interpolated values. A dash means that configuration has no original knot at that Mach value; the evaluator interpolates its own grid. The full-stack values through Mach 7 are digitized from Pezent's Figure 2.13(b) [7]; Mach 8/10 extensions and the three other curves are A-class. Coefficients must retain their stated reference areas. Shape transfer to the CRS-10 loaded-Dragon envelope is unvalidated.

Table 19 Complete stored zero-angle axial coefficient knots; return and landing share one column.

Mach	Full stack	Booster alone	Return / landing	Declared upper (inactive)
0	0.1416	0.42	0.67	0.32
0.05	0.1289	—	—	—
0.1	0.1221	—	—	—
0.2	0.1173	—	—	—
0.4	0.1124	—	—	—
0.6	0.11	—	—	—
0.8	0.1075	0.41	0.66	0.31
0.85	—	0.4	0.65	0.3
0.9	0.1075	—	—	—
0.92	0.1552	—	—	—
0.94	0.2631	—	—	—
0.96	0.4257	—	—	—
0.98	0.5954	—	—	—
1	0.7655	0.72	0.97	0.62
1.02	0.9342	—	—	—
1.04	1.0799	—	—	—
1.05	—	0.94	1.19	0.84
1.06	1.1615	—	—	—
1.1	1.1562	0.92	1.17	0.82
1.2	1.1262	—	—	—
1.3	—	0.84	1.09	0.74
1.4	1.0732	—	—	—
1.6	1.032	0.74	0.99	0.66
2	0.9739	0.68	0.93	0.6
2.5	0.9299	—	—	—
3	0.9007	0.56	0.81	0.48
4	0.8634	0.5	0.75	0.42
5	0.8423	0.46	0.71	0.38
6	0.8253	—	—	—
7	0.8131	0.42	0.67	0.36
8	0.79	0.41	0.65	0.35
10	0.75	0.4	0.63	0.34

A.3 Mission, control, sensor and filter settings

All settings in Tables 20 and 21 are A-class scenario choices unless the table explicitly identifies a published timing anchor. They describe the executed full case after `recovery_plan` transforms clock-based entry/landing references into state-triggered conditions. The JSON's preflight entry/landing times are comparison markers, not the active restart clock.

Table 20 Complete nominal mission/control setting groups and numerical criteria.

Setting family	Value / active condition	Meaning / limitation
Initial structural base	$\boldsymbol{r}_O^B = [6371000, 0, 0]^\mathsf{T}$ m; $\boldsymbol{v}_O^B = \mathbf{0}$; $\boldsymbol{\omega}^B = \mathbf{0}$	Base initially stationary relative to Earth. CG velocity includes migration.
Initial attitude / engines	$\boldsymbol{q}_{BE} = [\sqrt{1/2}, 0, -\sqrt{1/2}, 0]^\mathsf{T}$; nine engines at throttle 1	Hamilton scalar-first; Body +x up.
Ascent pitch / PD gains	Start 8 s; $\tau = 100$ s; tilt cap 80°; $k_p = 2\,\mathrm{s}^{-2}$, $k_d = 2.5\,\mathrm{s}^{-1}$	Unlimited ideal ascent couple. Stored heading 45° is ignored by the active eastward target construction.
Shutdown / separation	Ramp 140–141 s; separation 144 s; zero flow/flow derivative and engines off	Published approximate anchors [1]; one-second quintic envelope and no-impulse map assumed.
Return timing / engines	Guidance 0.2 s; ramps 1 s; boostback/entry indices $(0, 1, 5)$; landing (0)	Allowed landing guidance interval is 0.01–0.2 s.
Fuel allocations / protected reserve	Boostback 38000; entry 16000; landing 10000; reserve 2000 kg	Unused boostback/entry allocation can transfer; limiting-tank residual protected separately.
Boostback start / end design	161 s; throttle 1; window 10–60 s; predicted miss target 500 m	Fuel/duration cap can terminate before miss target. 500 kg anticipatory shutdown margin.
Boostback predictor / steering	Prediction 0.5 s steps, refreshed every 1 s; time gain 8 s; up bias $0.8\,\mathrm{m\,s}^{-2}$	SI numerical heuristic; $\boldsymbol{e}_{\mathrm{miss}}$ is predicted horizontal miss, t_f predicted time to impact; denominator floor $1\,\mathrm{s}^2$; normalized direction.
Entry trigger / burn	Descending through 45000 m; throttle 0.8; end speed $500\,\mathrm{m\,s}^{-1}$	Fuel-cap shutdown uses 500 kg anticipation.
Restart readiness	Separated, engines off, zero flow, fuel available; pointing error $< 25°$	Engine health, settling and ignition success assumed, not modeled.
Landing ignition prediction	0.1 s stopping-predictor step; factor 1.05; clearance allowance 1500 m	Reduced prediction, not observed CRS-10 ignition state.
Landing target	North −14000 m; east 6000 m; projected onto the ground sphere	Rounded assumed target, not surveyed LZ-1 coordinates.
Contact geometry	Four leg tips, 1 m below base, radius 3.5 m	Lowest tip determines first sphere crossing; no contact-force model.
Return couple / command limits	Powered 2 MN m; coast 200 kN m; $0.12\,\mathrm{rad\,s}^{-2}$, $0.22\,\mathrm{rad\,s}^{-1}$	Residual ideal-couple limits; acceleration cap is a command norm, rate clip is per desired-rate axis. Not measured-state bounds.
Return attitude gains	$k_p = 1\,\mathrm{s}^{-1}$; $k_d = 2\,\mathrm{s}^{-1}$	Cascaded direction-to-rate and rate-to-acceleration gains.
Landing attitude gains / tilt	$k_p = 1.5\,\mathrm{s}^{-1}$; $k_d = 6\,\mathrm{s}^{-1}$; tilt 20°, terminal 4°	Terminal hold below 20 m; height-dependent taper uses 0.02°/m.
Landing velocity / lateral gains	$v_t = -0.5\,\mathrm{m\,s}^{-1}$; $k_p = 0.0576\,\mathrm{s}^{-2}$; $k_d = 0.48\,\mathrm{s}^{-1}$	Vertical clearance feedback and lateral target feedback.
Landing criteria	$\lvert v_{\mathrm{vert}}\rvert \le 2$, $v_{\mathrm{horiz}} \le 1\,\mathrm{m\,s}^{-1}$; tilt ≤ 5°; rate $\le 1°\,\mathrm{s}^{-1}$; miss ≤ 10 m	Contact and deployed legs required; evaluated before contact-state reset (none is applied).
Upper ignition / cutoff / ramp	152/545/1 s; throttle 0.95	Timing anchors [1]; operating commands assumed.
Upper target / feedback times	200000 m; height time 80 s; velocity time 40 s	Project altitude/radial-speed feedback; not an insertion reconstruction.

Continued on next page

Table 20 continued

Setting family	Value / active condition	Meaning / limitation
Upper PD / drag / step	$2\,\mathrm{s}^{-2}$, $2.5\,\mathrm{s}^{-1}$; $C_D = 0.3$; $\Delta t_{\max} = 0.05\,\mathrm{s}$	Uses stage cross-section and a separate drag provider; no upper navigation instance.
Physical integration / output	R1 RK4 $\Delta t_{\max} = 0.01\,\mathrm{s}$; output 0.1 s; requested end 600 s	Booster stops at accepted contact; upper continues on the same mission clock.
Contact refinement / tolerances	At most 35 bisections; time bracket 10^{-9} s; separation relative 10^{-10}	Structural/CG continuity 10^{-8} m or $\mathrm{m\,s}^{-1}$; attitude/rate 10^{-12} in their norms.

Table 21 Nominal sensor and estimator setting register (all A-class research assumptions).

Setting family	Adopted value	Interpretation
IMU station / rate	$[38, 0, 0]^\mathsf{T}$ m / 100 Hz	Fixed structural station; not the CG.
Accelerometer white noise / bias	$0.01\,\mathrm{m\,s}^{-2}$ per axis / $[0.05, -0.03, 0.02]^\mathsf{T}\,\mathrm{m\,s}^{-2}$	Per-sample standard deviation / constant Body bias.
Accelerometer scale / bias walk	$[10^{-3}, 10^{-3}, 10^{-3}]$ / $10^{-4}\,\mathrm{m\,s}^{-5/2}$	Walk increment standard deviation is density times $\sqrt{\Delta t}$.
Gyro white noise / bias	$10^{-4}\,\mathrm{rad\,s}^{-1}$ per axis / $[0, 0, 10^{-4}]^\mathsf{T}\,\mathrm{rad\,s}^{-1}$	Relative-to-E angular-rate convention.
Gyro bias walk / scale	$10^{-7}\,\mathrm{rad\,s}^{-3/2}$ / zero true scale factors	Calibration states estimated despite zero nominal true scale errors.
Cross-axis / temperature terms	Accelerometer/gyro misalignment zero; all temperature coefficients zero; 20°C constant	Supported error terms switched off; no thermal experiment.
GNSS station / rate / latency	$[41, 0, 0]^\mathsf{T}$ m / 10 Hz / 0.1 s	Delayed antenna position/velocity with lever arms.
GNSS white position / velocity	$[1.5, 1.5, 2.5]^\mathsf{T}$ m / $[0.1, 0.1, 0.1]^\mathsf{T}\,\mathrm{m\,s}^{-1}$	Per-axis standard deviations.
GNSS Gauss–Markov component	$[0.6, 0.6, 1.0]^\mathsf{T}$ m; correlation time 300 s	Stationary standard deviation; matching bias states and exact discrete Gauss–Markov process in the filter.
GNSS acquisition / faults	First fix 0 s; dropout 0; outage list empty	Nominally all emitted GNSS measurements valid.
LiDAR station / axis / rate	$[0.5, 0, 0]^\mathsf{T}$ m / $[-1, 0, 0]^\mathsf{T}$ / 10 Hz	Analytic ray–sphere intersection; same Earth sphere.
LiDAR range / noise / faults	1–1800 m; $\sigma_R = 0.05\,\mathrm{m} + 10^{-4}R$; bias/scale/dropout 0	Geometry rejects out-of-envelope or nonintersecting samples.
Star boresight / reference axis	$[0.5, \sqrt{3}/2, 0]^\mathsf{T}$ / $[1, 0, 0]^\mathsf{T}$	Sensor +z is boresight; mounting quaternion computed from these axes.
Star rate / latency / noise	4 Hz / 0.1 s / 5/40 arcsec	Cross-boresight / boresight-roll standard deviation.
Star visibility constraints	Rate $\le 2^\circ\,\mathrm{s}^{-1}$; altitude ≥ 60000 m; Earth-limb exclusion 20°	Physical validity does not guarantee filter acceptance.
Star other settings / gate	Mount error 0; first fix 0; dropout 0; outages empty; NIS gate 16.27	Sun exclusion 40° is inactive because no Sun direction is configured.
Filter state / initialization	31 nominal scalars; 30 covariance coordinates; $\delta \boldsymbol{p}_0 = [1, -1, 0.5]^\mathsf{T}$ m	Position perturbation; zero velocity/mean attitude perturbations; bias/calibration means zero.
Initial standard deviations	Position 2 m; velocity $0.05\,\mathrm{m\,s}^{-1}$; accel bias $0.1\,\mathrm{m\,s}^{-2}$	Per axis; diagonal initial covariance.

Continued on next page

Table 21 continued

Setting family	Adopted value	Interpretation
Other initial standard deviations	Attitude 0.01 rad; gyro bias 10^{-3} rad s^{-1}; scale 10^{-3}; misalignment 10^{-3} rad	Three gyro scale and six off-diagonal terms; accel scale prior 10^{-3} per axis; GNSS bias prior equals stationary deviations.
Filter process settings	$S_a = (0.01)^2/100$; $S_g = (10^{-4})^2/100$; scale/misalignment walk 0	Implementation PSD convention; bias-walk densities match the IMU. Estimated attitude, gyro calibration, accelerometer scale and colored GNSS bias enabled.
Random streams	R1 seed 1; IMU/GNSS/LiDAR/star seeds 1/2/3/4	Independent seeded generators; no trial-step random draws.

A.4 Variables, units and reference points

Table 22 resolves notation reused across the model and results. Context distinguishes the air gas constant R from LiDAR range R and measurement covariance $\mathbf{R}$; the Earth radius is always R_E. The filter position is not the physical CG position. Superscripts B and E specify components, not a different physical point.

Table 22 Principal variables and units; derivative units follow from elapsed time in seconds.

Symbol	Unit	Definition
$t, \Delta t, N$	s, s, dimensionless	Elapsed mission time, integration/sample interval, number of comparison samples.
B, E, I	frames	Current Body, launch-aligned Earth-fixed frame, and initially coincident inertial frame.
$O, C, \boldsymbol{s}, \boldsymbol{c}, \boldsymbol{d}$	points / m	Structural base, moving CG, station, CG station, station-minus-CG arm.
$\boldsymbol{r}_C^B, \boldsymbol{v}_C^B$	m, m s^{-1}	Earth-centre-to-CG position and Earth-relative CG velocity in current Body components.
$\boldsymbol{p}, \boldsymbol{v}$	m, m s^{-1}	Filter IMU-station position/velocity in E; hats denote estimates.
$\boldsymbol{q}_{BE}, \mathbf{C}_{BE}$	dimensionless	Scalar-first Hamilton quaternion and rotation mapping Body components into E.
$\boldsymbol{\omega}, \boldsymbol{\alpha}, \boldsymbol{\Omega}^E$	rad s^{-1}, rad s^{-2}, rad s^{-1}	Modeled relative Body rate, angular acceleration, and Earth-rate vector.
$m, m_i, m_{p,k}, \mathbf{J}_C$	kg, kg, kg, kg m^2	Vehicle/component mass, stage-indexed propellant, CG inertia tensor.
Q, Q_f, Q_o, μ_j	kg s^{-1}	Positive outward flow and fuel/oxidizer parts; signed removal $\mu_j = -Q_j$.
$\mathbf{G}_j, \mathbf{S}(\boldsymbol{d})$	m^2	Outlet gyration tensor and parallel-axis tensor per unit mass.
$T, u, I_{\mathrm{sp}}, A_e$	N, dimensionless, s, m^2	Engine thrust, throttle, specific impulse, exit area.
$\boldsymbol{F}, \boldsymbol{M}_C, \boldsymbol{g}, \boldsymbol{a}_{\mathrm{rot}}$	N, N m, m s^{-2}, m s^{-2}	Nongravitational resultant, CG moment, central gravity and frame acceleration.
$\rho, p, T_{\mathrm{air}}, a, \bar{q}, M, \alpha$	kg m^{-3}, Pa, K, m s^{-1}, Pa, –, rad	Air density, pressure, temperature, sound speed, dynamic pressure, Mach and total flow angle.
$C_{A0}, C_{N\alpha}, C_{Dc}, c_m, c_l$	–, rad^{-1}, –, –, –	Axial curve, normal slope, crossflow coefficient and pitch/yaw/roll damping coefficients.
$\boldsymbol{f}, \boldsymbol{b}_a, \boldsymbol{b}_g, \boldsymbol{s}_f, \boldsymbol{m}$	m s^{-2}, m s^{-2}, rad s^{-1}, –, rad	Specific force, accel/gyro biases, gyro scale and off-diagonal small-angle terms.

Continued on next page

Table 22 continued

Symbol	Unit	Definition
$\boldsymbol{s}_a, \boldsymbol{b}_G$	–, m	Accelerometer scale factors and colored GNSS position bias in E.
$\delta\boldsymbol{x}, \mathbf{P}, \mathbf{F}_k, \mathbf{Q}_k$	block-dependent	Error state, its covariance, discrete transition and process covariance.
$\boldsymbol{\nu}, \mathbf{H}, \mathbf{R}, \mathbf{S}, \mathbf{K}$	measurement-dependent	Innovation, measurement Jacobian/covariance, innovation covariance and Kalman gain. This $\mathbf{S}$ differs from the parallel-axis function by context.
e_q, σ, NIS	rad or deg, state-dependent, –	Shortest attitude-error angle, reported standard deviation and normalized innovation squared.

A.5 Reproduction, audit records and source excerpts

The source repository is https://github.com/Mimilv1/Falcon9-simulation.

R1 is `results/final-submission-20261110-nav30`, generated at runtime revision `2d89a886a761`; all 31 runtime/data fingerprints match the reviewed source. The archive also retains the paired earlier filter baseline and R2, the joint step/IMU-clock variant. The environment is Python 3.12.14, NumPy 2.3.5 and Matplotlib 3.10.7; package versions are pinned in `requirements.txt`. Seed equality alone does not specify floating-point/library behavior across platforms.

The tracked `report/reproduction_data.zip` contains nominal truth, both trajectories, raw measurements, estimates, full covariance, innovations, events and summaries, plus the compared baseline/variant records. Its SHA-256 and each member fingerprint are in `report/figure_sources.json`. Extraction checks paths and hashes and refuses to overwrite differing files. From a fresh checkout, the following commands unpack the records, regenerate the report's result/model panels and numeric macros, then rerun the independent fixtures:

```
python -m pip install -r requirements.txt
python report/reproduce.py --unpack --validate
python -m unittest discover -s tests -q
python -m unittest discover -s Environment -p "test*.py" -q
```

The extracted directory is `results/submission-reproduction`. Published photograph/press-kit crops and the unchanged environment/aerodynamic illustrations remain supplied assets with attribution; the script regenerates all reported flight/navigation panels, the architecture, mass sweep and mass-code excerpt. Figure fingerprints allow comparison with the supplied versions in the pinned environment. Original runtime byte hashes are preserved; separately recorded LF-normalized source hashes distinguish Git line-ending conversion from content changes. Archive records retain exact byte checks. An actual source change invalidates the nominal check rather than silently authenticating a new run as R1.

A new physical integration with R1 settings is requested using:

```
python main.py --case full --no-show --seed 1 --dt 0.01 --imu-rate 100
```

The default creates a new timestamped directory; an explicit `--output` path must not exist. The full case requests 600 s, with booster output ending at contact and the upper child continuing. The ascent-only case requests 160 s. Standard simulator plots can be reopened after unpacking:

```
python main.py --plot-only results/submission-reproduction/nominal --no-show
```

Independent evidence is recorded in `report/navigation_validation.json`. The accelerometer fixture uses 180 s at 20 Hz, seed 11, three-axis varying force, injected scale $[0.005, -0.003, 0.002]$ and bias $[0.01, -0.02, 0.005]^{\mathsf{T}}\,\mathrm{m\,s^{-2}}$. Its scale prior is 0.02; aiding at 2 Hz uses $0.01\,\mathrm{m}/0.005\,\mathrm{m\,s^{-1}}$ white noise, no latency or colored bias, to isolate calibration.

The gyro fixture uses 450 s at 50 Hz, seed 12, fixed 150 km position and analytic attitude $\boldsymbol{q}_0 \otimes q_x(0.3\sin 0.031t) \otimes q_y(0.4\sin 0.047t) \otimes q_z(0.35\sin 0.063t)$; q_i is a rotation about axis i and the coefficients

use radians/seconds. Injected gyro scale is $[0.002, -0.0015, 0.001]$, bias $[0.0005, -0.0003, 0.0002]^\mathsf{T}\,\mathrm{rad\,s^{-1}}$, and cross-axis terms in xy, xz, yx, yz, zx, zy order are $[0.001, -0.0008, 0.0006, 0.0012, -0.001, 0.0005]$. Initial attitude error is $[0.003, -0.002, 0.004]^\mathsf{T}$ rad; gyro scale/cross-axis priors are 0.003. Its star boresight is Body $+x$; the nominal flight retains the 60-degree mounting. Rate RMS uses the final 100 s. This fixture tests calibration under deliberate excitation, not the CRS-10 mounting/trajectory.

The position-consistency fixture uses 20 independent prior/noise draws, 120 s at 20 Hz, fixed attitude and constant velocity $[1, -0.2, 0.1]^\mathsf{T}\,\mathrm{m\,s^{-1}}$. Initial position/velocity/bias deviations are 2 m, $0.1\,\mathrm{m\,s^{-1}}$ and $0.03\,\mathrm{m\,s^{-2}}$. GNSS is 2 Hz with 30 s correlation and zero latency; other GNSS noise levels retain sensor defaults. These accelerated correlation/sampling settings are fixture choices, distinct from nominal 10 Hz/300 s. Exactly one final position error per trial enters ANEES. The analytic-spin RK4 test uses $[0, 0, 0.4]^\mathsf{T}\,\mathrm{rad\,s^{-1}}$ for 10 s and no sensor/guidance clocks. Full settings, error norms, tolerances and traces remain executable and archived.

Table 23 describes the saved records; Table 24 gives the R1 fingerprints used to regenerate the figures. Hash equality authenticates the compared files, not the validity of their physical assumptions. The report is compiled from `report/main.tex` with its supplied class support; each chapter uses `\include`.

Table 23 Run-output contract.

Record	Contents
`truth.csv`	Accepted stack/booster output epochs: CG state, loads, mass/CG/inertia, atmospheric and trajectory values.
`stage2_truth.csv`	Upper-child state from the inherited separation state through the requested end.
`measurements.csv`	Sensor values, validity, effective exposure epochs/latencies and star blinding reasons; IMU truth channels are evaluation-only.
`navigation.csv`	IMU-station estimates, attitude/calibration errors, 30 standard deviations, innovation and aiding-acceptance diagnostics.
`navigation_covariance.npz`	Every saved navigation epoch and full 30×30 error covariance.
`startracker_innovations.csv`	Every valid star exposure: NIS, acceptance flag and residual angle.
`events.json`	Accepted events, separation audit and contact criteria; upper-child events.
`stage2_events.json`	
`event_timeline.csv`	Combined mission-clock event sequence for both vehicles.
`validation.csv`	Retained/interpolated ascent comparison values and residuals.
`summary.json`	Run status, plan, seed, sensor specifications, input hashes and aggregate results.

Table 24 R1 saved-record SHA-256 fingerprints.

Record	SHA-256
`summary.json`	`c7282111f5dbf4ae5bd8a7103857d51c6d8a12eb8f35ecc403d35715f5ed304c`
`truth.csv`	`49b8cdac785c58eef19e438a172cddf2a84eb1bc6172c367e39121ef4e92fd44`
`stage2_truth.csv`	`ac510163a9f32ff8743a14f894d879df556d3790ec765f27e1ca71d58b2a252d`
`navigation.csv`	`451b6e124cb40bb773c01a76e07029e1e82c556b5d6a46bc757e2f9179cb90ba`
`measurements.csv`	`4493598f290ccae7b36d1bb98eeb4809d732f20090ae00a525ce124222b1983c`
`events.json`	`8411b6cfcc5c29e54e4aafc67b4592e735401f6ebfdc172a006e52b77c26a311`
`validation.csv`	`4897c1bfcf9899e8d0afbef02235b2e884c241d98e430bca01d9c543a2e6666e`

Figures 13, 14 and 15 print selected unmodified source excerpts. They expose the load-to-motion and child-state interfaces; the complete modules remain the executable code deliverable. Source ranges and full-file hashes are stored with the figure provenance. The mass/inertia summation already printed in Figure 3 is not repeated here.

```
force = np.zeros(3)
moment = vector3(free_moment_body_nm).copy()
for applied in applied_forces:
    force += applied.force_body_n
    moment += np.cross(
        properties.arm_from_station(applied.station_body_m), applied.force_body_n
    )
return Loads(
    force,
    moment,
    gravity_body_mps2,
    consumption_kgps,
    tuple(exhaust_ports),
    frame_acceleration_body_mps2,
)
```

Figure 13 Force aggregation and CG moments in `load_interface.py`. A station arm enters the cross product once, while gravity and frame accelerations remain separate load fields. Source lines 27–41.

```
acceleration = (
    loads.force_body_n / mass
    + loads.gravity_body_mps2
    + loads.frame_acceleration_body_mps2
    + correction / mass
)
angular = np.linalg.solve(
    properties.inertia_body_kgm2,
    loads.moment_body_nm
    - np.cross(omega, properties.inertia_body_kgm2 @ omega)
    - properties.inertia_rate_body_kgm2ps @ omega
    + angular_flux @ omega,
)
return acceleration, angular
```

Figure 14 Acceleration closure in `dynamics.py` after the outflow correction and angular-flux tensor are assembled, implementing Equations (7)–(8). Exhaust reaction is not added twice. Source lines 163–176.

```
def separate(self, index=1):
    if not 0 < index < len(self.stages):
        raise ValueError("index out of range")
    cut = self.stages[index].offset
    r_body = np.array([cut, 0.0, 0.0])
    lower, upper = deepcopy(self), deepcopy(self)
    lower.stages = lower.stages[:index]
    lower.payload_mass = 0.0
    upper.stages = upper.stages[index:]
    for stage in upper.stages:
        stage.offset -= cut
    upper.datum_shift_m += cut
    lower.navigation = upper.navigation = None
    lower.navigation_imu_station_body_m = upper.navigation_imu_station_body_m = None
    upper.position = self.position + r_body
    upper.velocity = self.velocity + np.cross(self.omega, r_body)
    return lower, upper
```

Figure 15 Structural-base child mapping in `VehicleData.py`. The upper datum shifts by the cut station and its velocity inherits the rotational cross term. `staging.py` separately guards zero flow and audits conservation. Source lines 168–184.

A.6 Supplementary results and interpretation

Figure 16 supplements the main results with changing properties and sensor validity. Table 25 distinguishes emitted, valid and used samples; valid star visibility does not establish accepted aiding. Table 26 retains the

joint step/IMU-rate contact diagnostic: both cases pass, but it is neither an isolated convergence-order study nor an ensemble robustness result.

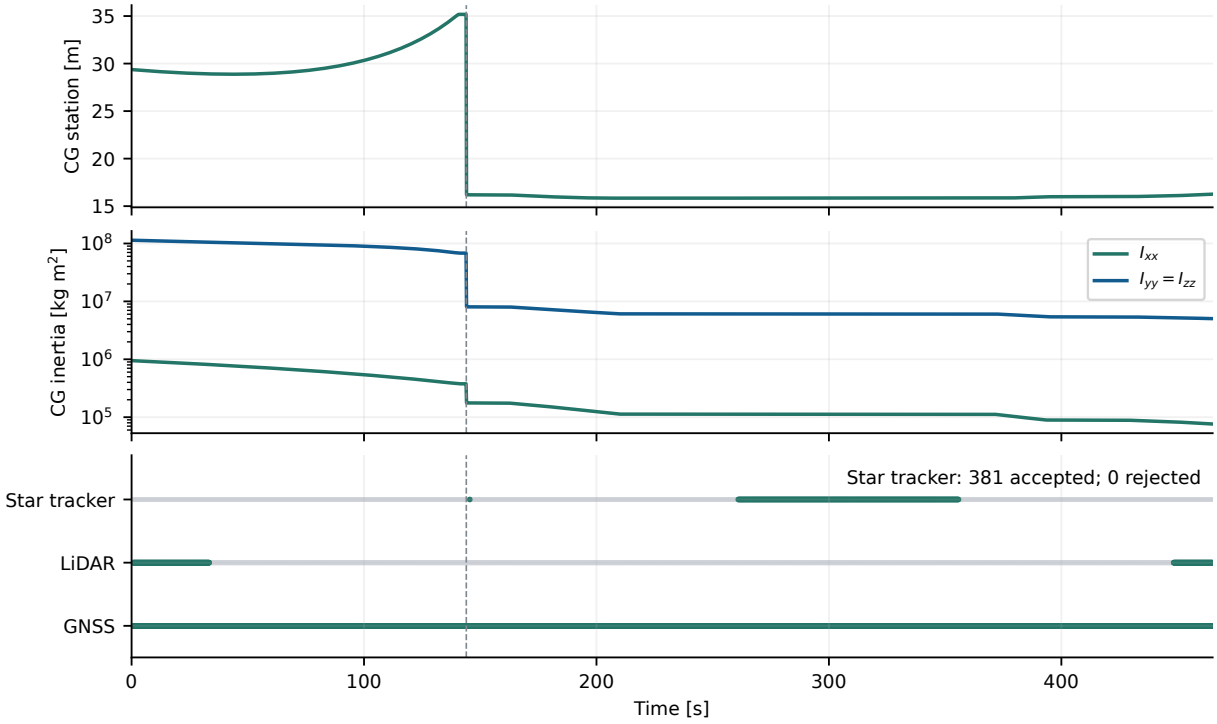


Figure 16 R1 properties and sensor validity. The dashed line marks separation at 144 s: the CG/inertia jump changes the tracked configuration, while structural-point states remain continuous. Green samples are valid and gray invalid. IMU validity is continuous and omitted from the panel. All 381 valid star exposures are accepted by the unchanged NIS gate.

Table 25 R1 sensor counts. The first IMU sample primes history; 46508 subsequent samples propagate the filter. Other used samples correct it.

Sensor	Emitted	Valid	Used
IMU	46509	46509	46509
GNSS	4651	4651	4651
LiDAR	4651	496	496
Star	1861	381	381

Table 26 Contact metrics under the two timing settings. Same target and acceptance limits.

Metric	R1	R2
Max. step [s]	0.01	0.05
IMU rate [Hz]	100	20
Contact time [s]	465.0827	465.1900
Vertical speed [m/s]	0.4975	0.5030
Horizontal speed [m/s]	0.9016	0.7404
Tilt [deg]	0.5349	1.8062
Angular rate [deg/s]	0.01289	0.11308
Target error [m]	3.3061	6.1705
Contact criteria	Pass	Pass